

# IS THERE A PLATONIC TREE?

## CALIBRATED CLASS GEOMETRY IN FOUNDATION MODELS

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### ABSTRACT

Foundation models are often said to converge toward shared representations. We ask this question of geometry: is a model’s arrangement of classes tree-like, and is that tree shared? Raw tree-likeness (Gromov  $\delta$ ) is confounded by dimension, class count and covariance, so we build a calibrated instrument that reads every score as an excess over a random cloud with the same dimension and spectrum. Across 12 vision backbones, 6 datasets and 16 text models the instrument gives a nuanced answer. Structure beyond the random baseline is nearly universal in vision (69 of 72 cells) and, in text, depends on the training recipe and emerges with scale; a star calibration shows that this structure is clustered rather than deep: models organize classes around hubs, with little hierarchy beyond, whether or not they were trained in hyperbolic space. The trees are moderately shared: a naive comparison of the models’ hierarchies isolates the self-supervised family as an island, an artifact of the clustering step; once controlled, every training recipe recovers the human taxonomy partially, more so with supervision. The same calibrated scores also indicate, with modest gains, where non-Euclidean readouts help. What foundation models share is clustered structure and a partial, supervision-dependent recovery of the human taxonomy — not one common tree.

## 1 INTRODUCTION

**Is there a Platonic tree?** Foundation models organize classes into calibrated, clustered structure and partially recover the human taxonomy, more so with supervision. They do not converge to one common tree.

figures/fig1\_concept.pdf (conceptual figure, drawn by the authors; same-size placeholder)

**Figure 1: Raw tree-likeness cannot tell a random cloud, a star and a tree apart; the calibrated instrument can.** A high-dimensional random cloud, clusters around a hub, and a nested hierarchy all give a low raw  $\delta$ . The excess over a spectrum-matched null separates the random cloud from the other two; only the matched-star test separates the star from the tree.

Semantic knowledge is hierarchical: instances belong to categories (a husky is a dog), categories to superordinate classes (dogs are mammals), and so on upward. The natural geometric home for such hierarchies is hyperbolic space (Nickel & Kiela, 2017; Sarkar, 2011), and a long line of work

imposes it at training time (Khrulkov et al., 2020; Chen et al., 2022; Desai et al., 2023). A complementary observation, latent hyperbolicity, holds that standard Euclidean models already organize concepts in a tree-like metric structure (Khrulkov et al., 2020; Yang et al., 2026). Is that structure hierarchical, and for whom?

The naïve measurement of this question is confounded. Gromov  $\delta$ -hyperbolicity (Gromov, 1987) assigns 0 to trees, but three artifacts push raw readings down without any hierarchy: high-dimensional clouds concentrate toward equidistance (a degenerate star tree), anisotropic spectra mimic low-dimensional behavior, and small class counts shrink the quadruple population. These artifacts are real: small GPT-2 models sit *at* their matched null despite low raw  $\delta$ , and classic sentence embedders share raw  $\delta$  with vision-contrastive models while carrying none of their genuine structure.

Our first contribution is an **instrument**: the exact estimator calibrated on reference geometries (trees, hyperbolic regions, spheres, Gaussians of growing dimension), with every reading expressed as an *excess* over a null cloud matched in dimension and covariance spectrum and ranked against 20 null replicates. With it we ask of geometry what convergence work asks of similarity (Huh et al., 2024; Gröger et al., 2026): what survives calibration, and is it shared — is there a Platonic tree?

**What survives is clustered form.** Structure beyond the matched null is nearly universal in vision (69 of 72 model $\times$ dataset cells) and emerges with recipe and scale in text. A star calibration shows what that structure is: classes organized around hubs, with depth beyond a matched star at most marginal, whether or not the model was trained in hyperbolic space.

**Sharing is graded, and its naive measurement misleads.** A naive tree-to-tree comparison isolates the label-free family as a categorical island, an artifact of the clustering step: once the configuration is controlled, ImageNet agreement rises from 0.03 to 0.38. Every training recipe recovers the human taxonomy partially, more so with supervision, and no configuration returns one common tree.

**Our contributions are:**

1. **A calibrated instrument for tree-likeness.** Estimator calibration on reference geometries, spectrum-matched nulls read as percentile ranks, and a class-count control. Identical raw readings receive opposite verdicts, so the raw scale is not interpretable.
2. **A calibrated census.** Genuine tree-like form in nearly every vision cell and, in text, form that depends on recipe and emerges with scale. A star calibration shows that the excess certifies clustering, not depth.
3. **The map of trees.** The naive comparison manufactures a categorical island. The controlled comparison shows graded, supervision-dependent recovery of the human taxonomy.
4. **A scoped corollary:** the same calibrated descriptors indicate where non-Euclidean readouts pay on prototype-based tasks.

## 2 RELATED WORK AND BACKGROUND

**Hyperbolic deep learning: imposing vs measuring.** Building on tree embeddability (Sarkar, 2011) and Poincaré embeddings (Nickel & Kiela, 2017; Ganea et al., 2018), hyperbolic methods span vision (Khrulkov et al., 2020; Ermolov et al., 2022; Atigh et al., 2022; Bdeir et al., 2024), vision-language (Desai et al., 2023; Pal et al., 2025), graphs (Chami et al., 2019), and language (Chen et al., 2022; He et al., 2026; Sinha et al., 2024). Latent hyperbolicity, low measured  $\delta$  in standard models, has been reported for CNN/ViT features (Khrulkov et al., 2020; Bdeir et al., 2024), token embeddings (Yang et al., 2026), and word vectors (Tifrea et al., 2019). Those readings are sample-level and uncalibrated; transplanted to sample-level features, our instrument finds excesses mostly within null noise (Appendix A.16), and our census is about inter-class (centroid) geometry.

**Geometric analyses of representations.** Representation similarity (Kornblith et al., 2019), intrinsic dimension (Ansuini et al., 2019; Pope et al., 2021), and neural collapse (Papayan et al., 2020) characterize Euclidean structure without testing tree-likeness. Estimating Gromov hyperbolicity on sampled quadruples has known bias and scale caveats (Fournier et al., 2015), which our calibration-plus-null design addresses operationally.

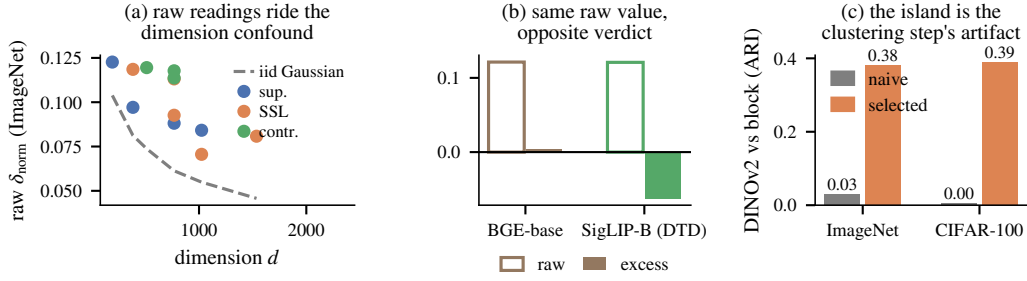


Figure 2: **Why calibration: raw  $\delta$  cannot be read on its own, and naive tree comparison manufactures islands.** (a) Raw  $\delta_{\text{norm}}$  rides the dimension confound: iid Gaussians (dashed) drift low as  $d$  grows, and the 12 backbones’ ImageNet readings sit on the same slope. (b) A text embedder (BGE-base) and a vision cell (SigLIP-B on DTD) share the same raw value with opposite null verdicts. (c) DINOv2-vs-block tree agreement before and after the configuration control of §5.

**Representational convergence.** The Platonic Representation Hypothesis (Huh et al., 2024) argues that models converge toward shared representations; Koepke et al. (2026) stress-test cross-modal convergence at scale, and Gröger et al. (2026) show that permutation-calibrated similarity removes most apparent global convergence while local neighborhood structure survives. Park et al. (2024) read hierarchical concept structure in LMs from the orthogonality of concept directions; our text census is a complementary, metric-level reading of the same question. We are the geometric complement of this program: the same calibration philosophy applied within models and, in §5, across them.

**Background.** For four points  $w, x, y, z$  of a metric space, sort the three pairing sums  $d(w, x) + d(y, z)$ ,  $d(w, y) + d(x, z)$ ,  $d(w, z) + d(x, y)$  as  $S_{\text{max}} \geq S_{\text{mid}} \geq S_{\text{min}}$ ; the four-point defect is  $(S_{\text{max}} - S_{\text{mid}})/2$  and  $\delta$  is its supremum. Trees have  $\delta = 0$ ; under this four-point definition the hyperbolic plane has  $\delta = \ln 2 \approx 0.693$  (Nica & Špakula, 2016), distinct from the thin-triangles constant  $\ln(1 + \sqrt{2}) \approx 0.881$ . We use the normalized  $\delta_{\text{norm}} = \delta_{\text{max}}/\text{diam}$  throughout: on this scale hyperbolic regions drift to 0 as they grow while spheres stay high, so the intuitive ordering trees < hyperbolic < spherical holds. Ollivier–Ricci curvature (Ollivier, 2009) on a kNN graph is our local companion: positive on tightly clustered neighborhoods, negative on tree-like bridges.

### 3 THE INSTRUMENT

This section builds the measurement layer in whose units every later claim is stated: raw  $\delta_{\text{norm}}$  is interpretable only against calibrated references and matched nulls.

**Classes are centroids; a cell is a (model, class set) pair.** Each frozen model is a fixed feature extractor; classes are represented by centroids (vision: the mean of each class’s cached training embeddings, 100 on ImageNet, the full cached split of 80–6,000 elsewhere; text: embeddings of “a photo of a {class}” over the 1000 ImageNet classes, plus the corpus variations of §4). The calibrated census covers 12 vision backbones  $\times$  6 datasets (a *cell* is one model  $\times$  dataset pair) and 16 text models (panel in Appendix A); the task grid covers the 10 vision backbones with cached per-dataset features (all but DINO-B and SigLIP-B). Estimator settings: the supremum four-point statistic is the pre-registered primary reading and the 99.9th percentile is reported only as robustness;  $5 \times 10^5$  quadruples per seed, 10 seeds (exact enumeration when  $C \leq 30$ ); ORC on  $k=10$  kNN graphs (ranking invariant for  $k \in \{5, \dots, 50\}$ ). On the transfer sets the excess is budget-stable from  $10^5$  quadruples and image-resampling moves it far less than its size; on ImageNet every sign holds at every budget but magnitudes drift by up to 0.02; a 99.9th-percentile variant preserves every sign (Appendix A.13).

**A low raw value is not evidence.** On reference geometries (Table 3, Appendix A.2) the estimator recovers the tree zero and the  $\ln 2$  four-point constant of  $\mathbb{H}^2$  ( $\delta_{\text{norm}} \rightarrow 0$  as hyperbolic regions grow),

spheres stay high (0.14–0.18), and iid Gaussians drift low as  $d$  grows (0.104  $\rightarrow$  0.046 from  $d=192$  to 1536): the dimension confound. A low raw value is not evidence.

**The reading is an excess over a matched null.** For every (model, class set) cell we synthesize null clouds with the same  $n$ ,  $d$ , mean and covariance *spectrum* (Gaussian coefficients recombined with the real singular values; alternative constructions in Appendix A.13), and report the **excess**  $\delta_{\text{norm}}^{\text{real}} - \delta_{\text{norm}}^{\text{null}}$  over 20 spectrum-null replicates, for  $\delta$  and  $\xi$  alike. The primary evidence is the *percentile rank*: how many of the 20 replicates lie above the real value (“below every replicate” = 20/20). Rank-based calibration is what gives permutation scores their finite-sample validity (Gröger et al., 2026, §5.1); a  $z$ -score on a handful of replicates is not rank-based, so  $z$  is kept only as a secondary summary in the appendix. The null is aggregation-aware in the sense of Gröger et al. (2026, §5.2): the same supremum over the same quadruple budget ( $5 \times 10^5$  per seed) is evaluated on every null replicate, so the max-statistic inflation is calibrated rather than ignored; the real cloud averages 10 quadruple seeds and each replicate 5, which affects variance, not bias. “Tree-like” below always means *negative excess*, and “genuine” means *beyond the spectrum-matched Gaussian null*, one hypothesis of “no hierarchy” (each null’s scope: Appendix A.1). Sign-positive excess means *less tree-like* than chance given the spectrum.

**$\xi$  reads norm structure, not curvature.** A parallelogram estimator  $\xi$  (midpoint defect of triangles) was intended as a curvature sign, but references with heterogeneous norms and no curvature drive it negative (Appendix A.19); we therefore report it only as a descriptor of norm structure, never as curvature.

**Class count is a measured variable, not a nuisance.** Because  $C$  sets the quadruple population, we sweep ImageNet subsets  $C \in \{10, \dots, 1000\}$  with two compositions: random (spanning the full hierarchy) and WordNet-coherent (siblings, effectively *flat*). Results in §4; the control makes “needs a real hierarchy” a measured variable.

**One parameter-free projection and two tasks serve the corollary.** The parameter-free map  $\phi(x) = \tanh(\frac{1}{2}\|s(x - \mu)\|) \cdot \frac{s(x - \mu)}{\|s(x - \mu)\|}$  centers features, scales the 95th-percentile norm to radius  $1/\sqrt{2}$ , and preserves directions; prototypes in the ball are means of projected points re-mapped through the origin. Tasks: nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes); metric arms: Euclidean (R), Poincaré (H), the radial map with Euclidean distances (RT), cosine (COS), and the L2-normalized stack. Changing the readout metric on the intact configuration, which is what the rule selects among, is distinct from deforming the configuration itself.

## 4 FINDINGS I: STRUCTURE BEYOND THE NULL

**Vision: structure beyond the null in 69 of 72 cells.** Table 1 summarizes the census. Sign: 69 of 72 cells have negative excess. The three sign-positive cells are DINOv2-S/B/G on ImageNet: above every null replicate, but by a small margin (excess  $\leq +0.012$ ), and below every replicate under the robustness statistic (Appendix A.13). Rank: 57 of 72 cells lie below every one of the 20 null replicates, 43 of 48 on the four hierarchical-label datasets (ImageNet, CIFAR-100, CIFAR-10, DTD). The flat datasets are sign-consistent (24/24) but sit near the estimator’s resolution limit at  $C=10$ , so only 14 of 24 clear every replicate (class-count control below; per-cell ranks in Appendix A.4). The magnitude, not the sign, of the small- $C$  excesses depends on how the null is built: an exact-sample-spectrum construction shrinks DINOv2’s CIFAR-100 excess while signs agree in 35/36 cells (Appendix A.13). Two ResNet-50 self-supervised models from the centroid store (Barlow Twins, BYOL) also sit below their null on ImageNet (Table 7), so the form is not a ViT artifact. On ImageNet, supervised ViTs sit at excess  $-0.019$  to  $-0.031$  and CLIP/SigLIP at  $-0.017$  to  $-0.027$ ; DINOv2-L is below every replicate on ImageNet, with a small excess ( $-0.008$ ), while within each transfer dataset the same family’s excess deepens with scale (S $\rightarrow$ G on CIFAR-10:  $-0.095 \rightarrow -0.155$ ; per-dataset trends in Figure 3b). Because excess magnitudes shrink with  $C$  (class-count control below), cross-dataset comparisons are not meaningful; all statements here are within-dataset trends. Under the supremum-robust p99.9 statistic the census itself is unchanged, 70/72 sign-negative and 42/48 hierarchical cells below every replicate, but the ImageNet ordering shifts: DINO/DINOv2 lie below every one of 20 null replicates while ViT-T’s supremum excess

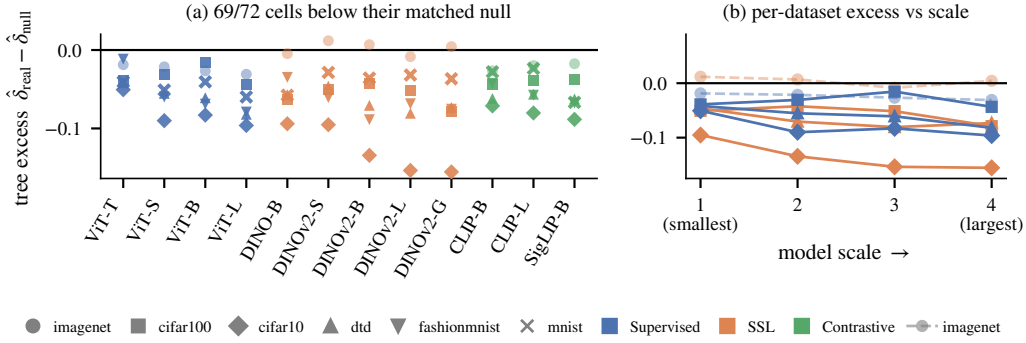


Figure 3: **Tree-like structure beyond the matched null is nearly universal in vision.** (a) Excess per vision cell (12 models  $\times$  6 datasets); the exceptions are DINOv2-S/B/G on ImageNet, above their null by less than the estimator noise. (b) Per-dataset excess vs scale: transfer-set excesses (solid) deepen with scale within each dataset, DINOv2’s ImageNet supremum excess (dashed) stays at null, and cross-dataset magnitudes are not comparable.

| model    | family | $\delta_{\text{norm}}$ IN | excess IN ( $r/20$ )   | exc. C100 | exc. C10 | $\rho_{\text{WN}}$ |
|----------|--------|---------------------------|------------------------|-----------|----------|--------------------|
| ViT-T    | sup.   | 0.123                     | −0.019 (20/20)         | −0.039    | −0.051   | +0.52              |
| ViT-S    | sup.   | 0.097                     | −0.021 (20/20)         | −0.031    | −0.090   | +0.50              |
| ViT-B    | sup.   | 0.088                     | −0.027 (20/20)         | −0.015    | −0.083   | +0.49              |
| ViT-L    | sup.   | 0.084                     | −0.031 (20/20)         | −0.043    | −0.096   | +0.53              |
| DINO-B   | SSL    | 0.113                     | −0.005 (16/20)         | −0.063    | −0.094   | +0.36              |
| DINOv2-S | SSL    | 0.119                     | +0.012 ( <b>0/20</b> ) | −0.051    | −0.095   | +0.22              |
| DINOv2-B | SSL    | 0.093                     | +0.007 ( <b>0/20</b> ) | −0.042    | −0.134   | +0.18              |
| DINOv2-L | SSL    | 0.071                     | −0.008 (20/20)         | −0.051    | −0.154   | +0.18              |
| DINOv2-G | SSL    | 0.081                     | +0.004 ( <b>0/20</b> ) | −0.079    | −0.155   | +0.20              |
| CLIP-B   | contr. | 0.120                     | −0.027 (20/20)         | −0.044    | −0.071   | +0.57              |
| CLIP-L   | contr. | 0.118                     | −0.020 (20/20)         | −0.038    | −0.080   | +0.59              |
| SigLIP-B | contr. | 0.114                     | −0.017 (20/20)         | −0.037    | −0.088   | +0.58              |

Table 1: **The calibrated census: genuine tree-like structure in nearly every vision cell.** One row per backbone (supervised / self-supervised / contrastive blocks).  $\delta_{\text{norm}}$ : raw ImageNet reading; excess:  $\delta_{\text{norm}}$  minus the spectrum-matched null mean, with the percentile rank  $r/20$  = how many of the 20 null replicates lie above the real value (20/20 = below every replicate; bold: sign-positive, i.e. less tree-like than the null);  $\rho_{\text{WN}}$ : Spearman correlation of inter-centroid and WordNet distances. Cross-dataset magnitudes are not comparable (§4). DTD, the flat datasets, the p99.9 robustness statistic and the task columns are in Appendix A (Table 5).

does not survive (Appendix A.13). Under the primary statistic DINOv2 on ImageNet therefore sits at its null; under the robustness statistic it lies below every replicate. We report both and do not adjudicate; whose tree it is (§5) does not depend on the choice.

**The excess certifies clustering, not depth.** The spectrum excess measures structure beyond second moments, and a pure star already has such structure: a synthetic 30-cluster star with no hierarchy scores  $\approx -0.11$ , as deep as any real cell (Appendix A.5). Depth is therefore tested separately, by comparing each backbone with a *matched star* that copies its hub radius, its within-superclass spread and its cluster sizes (Table 9). The matched-star test finds residual depth marginal: on ImageNet only DINOv2-L beats its star beyond the noise, and on CIFAR-100 every backbone is *less* tree-like than its star. A backbone *trained* in hyperbolic space is no exception: MERU (Desai et al., 2023) and its Euclidean CLIP twin show the same clustering excess and the same absence of residual depth, and MERU’s Lorentz-metric  $\hat{\delta}$  matches its Euclidean value (Appendix A.5). The excess measures how classes are arranged, not the geometry they are embedded in; the class-subsampling control below measures the hierarchy’s reach, and *whose* tree it is is the subject of §5.

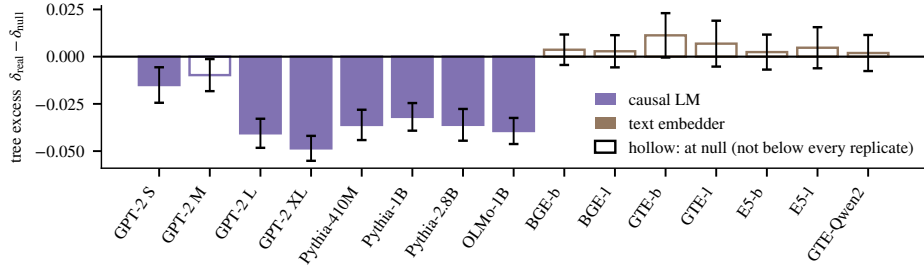


Figure 4: **In text, genuine form depends on recipe and emerges with scale.** Calibrated text census (padding-free extraction, 20 null replicates; error bars combine null and estimator s.d.; hollow bars sit at their null, i.e. not below every replicate; OLMo-7B, not re-extracted, omitted). Form emerges with scale in GPT-2, is constant in Pythia, and is absent in classic sentence embedders despite raw  $\delta$  equal to genuinely structured vision models.

**Text: the instrument reveals what raw  $\delta$  hides.** Within causal LMs, genuine form *emerges with scale* in GPT-2 (Figure 4): under padding-free extraction S and M sit at their null ( $-0.015/-0.010$ ) while L and XL lie below every replicate ( $-0.041/-0.049$ ; Table 12). Pythia carries genuine form at every scale, and OLMo at both scales tested. Extraction is part of the measurement: left-padded batching alters GPT-2’s hidden states and moves  $\hat{\delta}$  by up to 0.02, of the order of the marginal excesses (Table 13); the padding-free protocol removes this artifact, the scale trend survives it, and individual marginal cells (GPT-2 M) do not. The rising nulls track the effective rank of the centroid covariance, and genuineness is not a proxy for anisotropy (GTE-base has the highest rank and sits at its null); a PC-permutation null confirms every verdict. Classic sentence embedders (BGE/GTE/E5) sit at their nulls despite raw values indistinguishable from vision-contrastive models whose excess is genuine: identical raw values, opposite verdicts (Figure 2b). The same embedders show genuine excess ( $-0.020$  to  $-0.028$ ) on DBpedia, a real three-level text hierarchy: presence of form is a property of the (model, concept set) pair. The only LLM-backed embedder tested (GTE-Qwen2-1.5B) is not robust across replicate counts, and we do not build on it (Table 12). Causal-LM values are template-sensitive but the scale ordering is template-robust; per-template values, ranges and the padding-free re-run are in Appendix A.15.

**Hierarchy depth, not class count.** At fixed  $C$ , random ImageNet subsets, which span the full hierarchy, are consistently more tree-like and more exploitable than WordNet-coherent sibling subsets, an effectively flat concept set (DINOv2-L at  $C=50$ : excess  $-0.124$  vs  $-0.093$ ; Appendix A). Raw  $\delta_{\text{norm}}$  rises with  $C$ , so MNIST’s flat verdict is not a class-count artifact. DINOv2-G reaches its null at  $C \geq 500$  on ImageNet, reproducing the census exception.

**The form tracks training.** Two interventions move it (Figure 5; raw  $\delta$ , within architecture): fine-tuning on a task with no class hierarchy raises  $\delta$  by 19–91%, and  $\delta$  falls across transformer depth. Shuffling the embedding-to-label correspondence leaves it unchanged, as a null-invariant quantity should. Random-weight networks give higher raw  $\delta$ , but their features are near-Gaussian, so this is expected by construction and not counted as evidence; excess-based versions of these interventions need the stored point clouds and are left to future work. On matched architecture (ViT-B supervised/DINO/CLIP) all three objectives yield genuine form on most datasets, and the family separations emerge with scale rather than at fixed size.

**Negative curvature concentrates on bridges.** High mean ORC coexists with tree-likeness because they live on different edges: negative curvature concentrates on between-superclass bridges (about twice as often as within superclasses on ImageNet; Appendix A), while the mean measures within-cluster compactness. DINOv2 on ImageNet inverts the pattern, consistent with its at-null supremum reading, and recovers it on CIFAR-100.

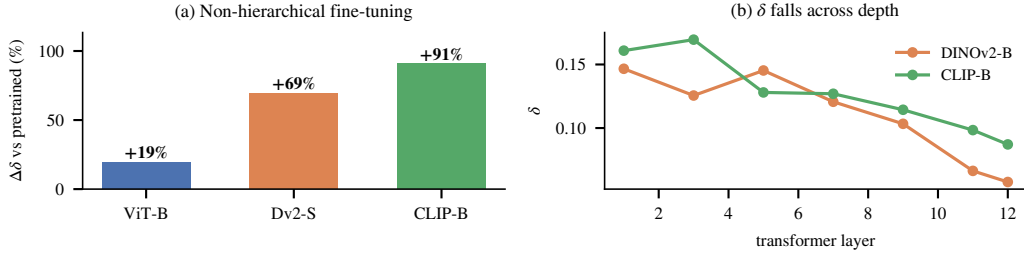


Figure 5: **The form is learned: training interventions move it.** Raw  $\delta$ , within architecture. (a) Non-hierarchical fine-tuning raises  $\delta$  19–91%. (b)  $\delta$  falls across transformer depth (layers  $\geq 1$ ).

## 5 FINDINGS II: SHARED CONTENT, FRAGILE MEASUREMENT

If the measured trees were a shared representation of reality, models should agree with each other and with human taxonomy. Whether they appear to depends on how the trees are measured: the naive-vs-calibrated gap is the paper’s thesis in miniature.

**The naive map manufactures an island.** We extract each model’s hierarchy explicitly (a dendrogram over the shared centroids) and compare hierarchies, not representations, by ARI between cut partitions (adjusted Rand index, chance-corrected partition agreement) and by the cophenetic correlation (CPCC) between tree and distances. Under the naive configuration, Euclidean distances with average linkage, the map (Figure 6) is spectacular: supervised ViTs and contrastive VLMs form one block (cross-family ARI up to 0.82), while the label-free DINOv2 family appears as an island (ARI  $\leq 0.06$  against every supervised model). The categorical island, however, is a measurement artifact. Its cuts are degenerate: DINOv2’s Euclidean average-linkage dendrograms chain (up to 97% of classes in one cluster), forcing ARI  $\approx 0$  by construction, and Euclidean distance is the metric our diagnostics flag as mismatched for this family (§4). A configuration must be selected by a criterion fixed *without reference to the agreements being compared*. Cophenetic fidelity alone re-selects the chaining configuration, so we fix: *highest CPCC among configurations whose reference cuts are non-degenerate for every model* (largest cluster  $\leq 50\%$ ); it selects cosine-average on ImageNet and cosine-complete on CIFAR-100 (Appendix A.8). The 0.5 degeneracy bound is the criterion’s one free parameter: across admissible bounds the ImageNet gap ranges from 0.13 to 0.38, never back to the categorical island ( $\leq 0.06$ ; Appendix A.8). Under the selected configurations (Figures 6b and 7b) the island shrinks from categorical to moderate (ImageNet 0.38 vs block 0.48). Across *all* non-degenerate configurations DINOv2 recovers the true 20 CIFAR-100 superclasses at ARI 0.2–0.5, reaching the supervised band only under cosine-Ward, the configuration where the CIFAR-100 island also closes fully (per-configuration values in Appendix A.8).

**The island is an artifact; sharing is graded.** Three statements hold across every non-degenerate configuration. The categorical island is an artifact. Label-free models recover the human taxonomy partially, and supervision deepens the resemblance: paired within each admissible configuration, the label-supervised ViT recovers more than each DINOv2 model in 11 of 12 comparisons (Appendix A.8). Tree-to-tree comparison itself is fragile to configuration. A moderate label-free gap persists on ImageNet under every configuration, consistent with the same reading: a real tree, least aligned with WordNet at the top.

**WordNet alignment follows supervision, not tree-likeness.** Spearman correlation between inter-centroid distances and WordNet tree distances on ImageNet: +0.57 to +0.59 for CLIP/SigLIP, +0.49 to +0.53 for supervised ViTs, +0.36 for DINO-B, and only +0.18 to +0.22 for DINOv2, the most tree-like family, with shuffle and Gaussian controls at  $\approx 0$  (Table 1). Cutting centroid dendrograms recovers the 20 true CIFAR-100 superclasses at ARI up to 0.61 (permuted labels  $\approx 0$ ). On HierarCaps (Alper & Averbuch-Elor, 2024) (1000 four-level caption chains), text encoders show the same reading with no class set: the projection radius orders levels within a chain, sibling triplets are resolved at 0.83–0.97 (chance 0.5), and the leaf excess repeats the text census (Table 38). Three controls bound circularity: WordNet-coherent groupings are more tree-like than random ones in 11/12

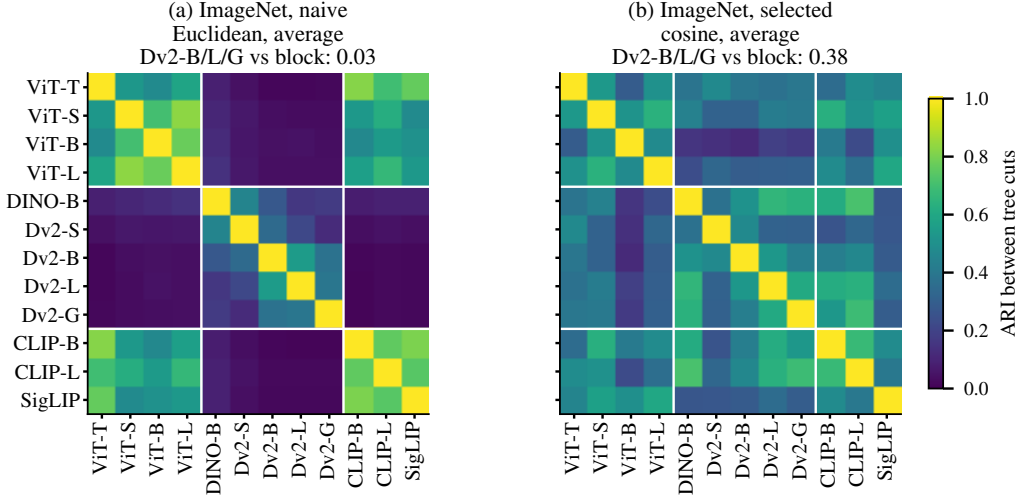


Figure 6: **The categorical island vanishes once the clustering step is controlled.** Pairwise ARI between dendrogram cuts of the 12 vision models on ImageNet (30 clusters): naive configuration (a: Euclidean, average linkage) vs criterion-selected (b: cosine, average), with the mean DINOv2-B/L/G-vs-block ARI above each panel. Per-configuration values in Appendix A.8; the CIFAR-100 pair, same reading, in Figure 7.

models; single-image distances still correlate with WordNet ( $\rho$  up to  $+0.26$ ); and the decoupling above is the opposite of label injection. DINOv2’s semantics concentrate in the angular component (sibling-triplet agreement 0.91 under cosine vs 0.71 Euclidean for DINOv2-L; Appendix A), which is also why Euclidean dendrograms chain for this family and produce the naive island. The census exception is consistent with this: whatever structure DINOv2 has on ImageNet (at null under the primary statistic, below every replicate under the robustness one), it is the least aligned with WordNet’s top level: the form-vs-content split.

**The recovery is not WordNet circularity.** ImageNet classes are WordNet synsets, so recovering WordNet there could be circular. Repeating the tree map on DBpedia Classes, a three-level ontology independent of WordNet (219 leaf classes; the three text embedders), closes this door: no configuration degenerates, the embedders recover the true 70-way level-2 partition at ARI 0.21–0.29 across all six configurations, and they agree with each other at cross-model ARI 0.65–0.82. The recovered non-WordNet hierarchy is real, shared, and configuration-robust; the fragility of §5 is a property of the (family, concept set) pair.

**Local agreement survives calibration.** We apply the permutation calibration of Gröger et al. (2026) as they define it: for each of the 66 model pairs,  $K=200$  permutations of the class correspondence give the null distribution, the threshold  $\tau_{0.05}$  is its  $[0.95(K+1)]$ -th order statistic, and the calibrated score is  $\max\{(s - \tau_{0.05})/(1 - \tau_{0.05}), 0\}$  (scalar calibration, no layer search). Local neighborhood agreement survives: mutual-kNN is 0.472 raw and 0.464 calibrated, with every pair at  $p = 1/201$ , and the null mean 0.010 equals the analytic chance level  $k/(n-1) = 10/999$ . Centroid CKA survives too (0.633 raw, 0.577 calibrated, null mean 0.108): centroids distill a genuinely shared relational base. The Poincaré reading improves neither agreement (kNN 0.472  $\rightarrow$  0.425; CKA 0.633  $\rightarrow$  0.640): hyperbolic distances buy no cross-model alignment, as expected, since  $\phi$  preserves directions and rescales norms monotonically (per-pair values and the null-centered variant in Table 21).

**Form and content anti-align.** The families with the most WordNet content (supervised ViTs, CLIP) carry the most fragile form, borne by the supremum statistic. The family with the most robust form (DINOv2) is the least aligned with WordNet’s top level.

## 6 COROLLARY: DIAGNOSING EXPLOITABILITY

Is there non-Euclidean inter-class structure worth exploiting for a frozen backbone, and which zero-cost metric collects it? The descriptors diagnose the first; label flatness and the training objective decide the second (full tables in Appendix A).

**$\delta_{\text{norm}}$  predicts the gain within datasets.** Within each dataset the Poincaré advantage tracks  $\delta_{\text{norm}}$  (NC  $r = -0.83$  on ImageNet), and the correlation survives partialing out dimension, baseline accuracy, family membership and spectral covariates (Appendix A.11). Without DINOv2 it holds on ImageNet and CIFAR-100 but not on CIFAR-10/DTT (Appendix A.11). Pooled cross-dataset correlations are family-driven and are not used as evidence; on flat-label datasets no metric helps. Headline gains carry paired per-episode confidence intervals and a McNemar test (Appendix A.11).

**The training objective picks the metric.** The identical radial map with Euclidean distances changes nothing, so the gain is the metric itself. The split follows the objective: cosine suffices for DINOv2, Poincaré  $\approx$  cosine for supervised ViTs, and Poincaré beats cosine for contrastive CLIP by up to +1.3pp even after L2 normalization (Figure 8 in Appendix A.11; per-cell values, the held-out selection and the policy comparison in Appendix A). The rule is read off the same three families it is scored on, so it describes these families rather than validating a policy.

**A diagnostic, not a policy.** Cosine is a safe default, and gating on  $\delta_{\text{norm}}$  is the weakest policy tested (Appendix A). Gains are modest and limited to prototype-based tasks; sample-level tasks never benefit, consistent with the weak sample-level excesses of Appendix A.16. On DBpedia, a three-level text hierarchy, the diagnostic predicts correctly on both sides: genuine excess, no metric gain.

## 7 DISCUSSION AND LIMITATIONS

**What the census establishes.** Foundation models organize classes into genuine, clustered structure and partially recover the human taxonomy, more so with supervision; they do not converge to one common tree. Form beyond matched nulls is widespread but not universal: near-universal in the vision backbones tested (ViTs and two ConvNets), recipe- and scale-dependent in text, always relative to a concept set. Its content is moderately shared and configuration-dependent: all recipes recover the human taxonomy partially under every non-degenerate configuration, the residual gap on ImageNet ranges from 0.13 to 0.38 with the comparison threshold, and the naive islands are a measurement artifact. Raw readings of  $\delta$ , of  $\xi$ , or of tree similarity are not interpretable without the instrument; excess measures existence while raw  $\delta_{\text{norm}}$  predicts exploitability, and the two dissociate in both directions (DINOv2 on ImageNet; the DBpedia embedders).

**Limitations.** (i) The spectrum excess certifies structure beyond second moments, star-like clustering included; against matched stars the residual depth is marginal (Appendix A.5), so the hierarchy’s reach rests on the class-subsampling control and its content on §5. (ii) The census is class-centroid geometry on six datasets; other concept sets may expose different structure. (iii) Causal-LM values (not the scale ordering) are template-conditioned. (iv) The objective–geometry link is correlational (triplet-bounded, not closed); the within-dataset  $\delta$ –gain prediction rests chiefly on DINOv2’s scale range (Appendix A.11). (v) Downstream scope is prototype-based metric tasks with modest gains. (vi) The parallelogram descriptor  $\xi$  tracks norm dispersion rather than curvature (Appendix A.19) and is reported only as such. (vii) A causal LM’s class representation depends on the extraction protocol as much as on the model: left-padded batching moves  $\hat{\delta}$  by up to 0.018, of the order of the L/XL excess.

## ETHICS STATEMENT

This work analyzes publicly released models and standard public datasets, involves no human subjects and no personal data, and proposes an analysis methodology and a metric-selection heuristic. We do not foresee ethical risks beyond those already associated with the underlying models and datasets.

## REPRODUCIBILITY STATEMENT

Every number in the paper is annotated in the source with the result file that produces it; all tables are generated by script from those files, and the full pipeline (feature extraction, diagnostics, nulls, calibration, tasks, statistics, figures) with fixed seeds will be released. Estimator settings are stated in §3. Computing  $\delta_{\text{norm}}$  on 1000 centroids takes seconds on CPU; the full task grid runs in under 30 minutes on one consumer GPU.

## THE USE OF LARGE LANGUAGE MODELS

LLM-based assistants were used, under continuous author direction and review, to draft and edit text, to write analysis scripts, and to run the experiment pipeline on the authors’ hardware. All experimental designs, results, verifications, and final claims were checked and approved by the authors, who take full responsibility for the content of this paper.

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## A FULL NUMERICAL TABLES

Every table is generated by script directly from the released per-cell result files; no value is hand-transcribed.

### A.1 WHAT EACH NULL PRESERVES

| null                                | preserves                               | destroys                                       |
|-------------------------------------|-----------------------------------------|------------------------------------------------|
| Gaussian, spectrum-matched          | $n, d$ , mean, covariance spectrum      | all higher-order structure                     |
| PC-permutation                      | each principal component’s marginal     | dependence across components                   |
| Cluster-preserving (tested, unused) | each cluster’s geometry, hub assignment | hub arrangement (invalid depth test: App. A.5) |
| Label shuffle (ablation)            | the point cloud itself                  | embedding–label correspondence                 |

Table 2: The nulls of the instrument and what each one preserves and destroys; “genuine” in the text always means beyond the first of these.

### A.2 ESTIMATOR CALIBRATION

| space                                                   | absolute $\delta$   | $\delta_{\text{norm}}$              |
|---------------------------------------------------------|---------------------|-------------------------------------|
| balanced binary tree (depth 10)                         | 0.000               | 0.000                               |
| $\mathbb{H}^2$ ( $K = -1$ ), region radius $R=2/4/8/16$ | 0.65/0.69/0.69/0.69 | 0.162/0.087/0.043/0.022             |
| uniform $S^{99}$ (chord / geodesic)                     | 0.25/0.37           | 0.143/0.179                         |
| iid Gaussian ( $n=1000$ ), $d=192, \dots, 1536$         | —                   | 0.104/0.081/0.074/0.061/0.055/0.046 |

Table 3: **A low raw  $\delta$  is not evidence of hierarchy.** Calibration on reference geometries: the estimator recovers the tree zero and the four-point constant of  $\mathbb{H}^2$ , spheres stay high, and iid Gaussians drift low as  $d$  grows, the dimension confound. Gaussian row from the released control file; tree/ $\mathbb{H}^2$ /sphere rows re-verified synthetically at every freeze.

### A.3 MODEL PANEL

Feature extraction: vision backbones use the encoder’s default pooled embedding (timm, num\_classes=0, i.e. CLS or global pool per architecture); causal LMs use the last-token hidden state, extracted one prompt at a time (padding-free: left-padded batching alters GPT-2’s hidden states, Table 13); text embedders use their recommended pooling (CLS for BGE, mean for GTE/E5, last token for GTE-Qwen2); CLIP/SigLIP text towers use their text encoders.

| type                    | model          | params |
|-------------------------|----------------|--------|
| Supervised ViT (vision) | ViT-T (i21k)   | 5M     |
| Supervised ViT (vision) | ViT-S (i21k)   | 22M    |
| Supervised ViT (vision) | ViT-B (i21k)   | 86M    |
| Supervised ViT (vision) | ViT-L (i21k)   | 307M   |
| SSL (vision)            | DINO-B         | 86M    |
| SSL (vision)            | DINOv2-S       | 22M    |
| SSL (vision)            | DINOv2-B       | 86M    |
| SSL (vision)            | DINOv2-L       | 307M   |
| SSL (vision)            | DINOv2-G       | 1.1B   |
| Contrastive (vision)    | CLIP-B         | 86M    |
| Contrastive (vision)    | CLIP-L         | 307M   |
| Contrastive (vision)    | SigLIP-B       | 86M    |
| Causal LM               | GPT-2 S        | 117M   |
| Causal LM               | GPT-2 M        | 345M   |
| Causal LM               | GPT-2 L        | 774M   |
| Causal LM               | GPT-2 XL       | 1.5B   |
| Causal LM               | Pythia-410M    | 410M   |
| Causal LM               | Pythia-1B      | 1.0B   |
| Causal LM               | Pythia-2.8B    | 2.8B   |
| Causal LM               | OLMo-1B        | 1.0B   |
| Causal LM               | OLMo-7B        | 7.0B   |
| Text embedder           | BGE-base       | 110M   |
| Text embedder           | BGE-large      | 335M   |
| Text embedder           | GTE-base       | 110M   |
| Text embedder           | GTE-large      | 335M   |
| Text embedder           | GTE-Qwen2-1.5B | 1.5B   |
| Text embedder           | E5-base        | 110M   |
| Text embedder           | E5-large       | 335M   |

Table 4: The full model panel: 12 vision backbones (all in the census; the 10 with cached per-dataset features, i.e. all but DINO-B and SigLIP-B, form the task grid), 9 causal LMs, and 7 text embedders.

#### A.4 VISION CENSUS: EXCESS AND SIGNIFICANCE PER CELL

| model    | p99.9 IN | exc. DTD | exc. FMNIST | exc. MNIST | best-R (pp) | metric |
|----------|----------|----------|-------------|------------|-------------|--------|
| ViT-T    | +0.007   | -0.042   | -0.012      | -0.041     | +0.58       | either |
| ViT-S    | -0.005   | -0.055   | -0.060      | -0.051     | +0.49       | either |
| ViT-B    | -0.006   | -0.061   | -0.069      | -0.041     | +0.51       | either |
| ViT-L    | -0.014   | -0.082   | -0.079      | -0.060     | +0.87       | either |
| DINO-B   | -0.025   | -0.057   | -0.035      | -0.058     | —           | —      |
| DINOv2-S | -0.022   | -0.046   | -0.061      | -0.029     | +0.68       | H      |
| DINOv2-B | -0.026   | -0.071   | -0.089      | -0.036     | +1.17       | cos    |
| DINOv2-L | -0.023   | -0.081   | -0.068      | -0.032     | +1.84       | cos    |
| DINOv2-G | -0.020   | -0.073   | -0.075      | -0.037     | +2.54       | cos    |
| CLIP-B   | -0.006   | -0.062   | -0.042      | -0.028     | +0.67       | H      |
| CLIP-L   | -0.011   | -0.056   | -0.058      | -0.023     | +0.65       | H      |
| SigLIP-B | -0.009   | -0.062   | -0.068      | -0.067     | —           | —      |

Table 5: Census columns moved out of Table 1. p99.9 IN: ImageNet excess under the supremum-robust statistic (20 replicates; Table 29); exc. DTD/FMNIST/MNIST: per-dataset excess on the remaining sets; best-R: few-shot advantage of the best zero-cost metric over Euclidean, mean over the four hierarchical datasets; metric: which of Poincaré (H) or cosine collects it (either: within 0.15pp; —: outside the 10-model task grid).

| model    | IN                   | C100           | C10           | DTD           | FMNIST        | MNIST         |
|----------|----------------------|----------------|---------------|---------------|---------------|---------------|
| ViT-T    | -0.021 (-2.7)        | -0.043 (-3.7)  | -0.059 (-2.0) | -0.037 (-3.9) | -0.022 (-0.7) | -0.049 (-1.4) |
| ViT-S    | -0.020 (-2.3)        | -0.033 (-4.0)  | -0.096 (-3.9) | -0.052 (-4.1) | -0.069 (-2.3) | -0.058 (-1.9) |
| ViT-B    | -0.028 (-3.0)        | -0.017 (-1.9)  | -0.090 (-3.1) | -0.058 (-4.8) | -0.076 (-2.6) | -0.043 (-1.4) |
| ViT-L    | -0.030 (-4.4)        | -0.048 (-6.0)  | -0.098 (-3.7) | -0.077 (-7.5) | -0.085 (-2.9) | -0.064 (-1.9) |
| DINO-B   | -0.005 (-0.6)        | -0.068 (-10.2) | -0.097 (-3.2) | -0.051 (-4.9) | -0.041 (-1.3) | -0.070 (-2.5) |
| DINOv2-S | <b>+0.012 (+1.1)</b> | -0.053 (-8.9)  | -0.095 (-2.2) | -0.040 (-4.2) | -0.068 (-2.3) | -0.036 (-1.2) |
| DINOv2-B | <b>+0.006 (+0.6)</b> | -0.043 (-11.3) | -0.135 (-2.7) | -0.068 (-6.6) | -0.095 (-3.3) | -0.045 (-1.4) |
| DINOv2-L | -0.008 (-1.1)        | -0.051 (-17.0) | -0.155 (-3.4) | -0.078 (-7.0) | -0.074 (-2.6) | -0.042 (-1.2) |
| DINOv2-G | <b>+0.004 (+0.5)</b> | -0.079 (-24.2) | -0.157 (-3.4) | -0.071 (-6.4) | -0.081 (-3.0) | -0.042 (-1.3) |
| CLIP-B   | -0.025 (-3.1)        | -0.047 (-5.8)  | -0.077 (-3.0) | -0.059 (-4.9) | -0.047 (-1.7) | -0.038 (-1.6) |
| CLIP-L   | -0.021 (-2.3)        | -0.043 (-5.8)  | -0.083 (-3.3) | -0.049 (-3.9) | -0.066 (-2.0) | -0.024 (-0.6) |
| SigLIP-B | -0.019 (-1.7)        | -0.042 (-6.6)  | -0.091 (-4.0) | -0.057 (-4.8) | -0.074 (-2.6) | -0.070 (-1.9) |

Table 6: Per-cell excess and  $z$ -score (in parentheses;  $z = (\delta_{\text{real}} - \bar{\delta}_{\text{null}})/\sqrt{\sigma_{\text{null}}^2 + \sigma_{\text{est}}^2}$ , 5 null replicates). Bold: the three sign-positive cells.

| model                    | $d$  | $\hat{\delta}$ | excess ( $z$ ) | reps above |
|--------------------------|------|----------------|----------------|------------|
| Barlow Twins (ResNet-50) | 2048 | 0.075          | -0.039 (-4.8)  | 20/20      |
| BYOL (ResNet-50)         | 2048 | 0.073          | -0.020 (-2.4)  | 20/20      |

Table 7: Backbones outside the ViT census, ImageNet centroid store: two ResNet-50 self-supervised models, under the spectrum null (20 replicates, percentile rank).

#### A.5 WHAT THE EXCESS CERTIFIES: SYNTHETIC CALIBRATION

| synthetic cloud ( $n=1000$ , $d=768$ ) | $\hat{\delta}$ | spectrum null A |        | hub-randomizing null B |        |
|----------------------------------------|----------------|-----------------|--------|------------------------|--------|
|                                        |                | Gaussian        | Haar   | Gaussian               | Haar   |
| 30-cluster star, tight (0.1)           | 0.055          | -0.115          | -0.115 | -0.126                 | -0.058 |
| 30-cluster star, mid (0.3)             | 0.059          | -0.104          | -0.103 | -0.114                 | -0.049 |
| 30-cluster star, loose (0.6)           | 0.061          | -0.084          | -0.082 | -0.090                 | -0.033 |
| 2-level hierarchy $6 \times 5$ (0.3)   | 0.044          | -0.143          | -0.143 | -0.113                 | -0.112 |

Table 8: Calibration of the nulls on synthetic clouds (excess, means over seeds; within/between-cluster noise ratio in parentheses). A pure star (Gaussian clusters around Gaussian centers, no hierarchy) sits far below the spectrum null A under either construction (Gaussian coefficients, the paper’s; Haar-rotated coefficients, exact sample spectrum): null A certifies clustering, not depth. The hub-randomizing null B, which keeps every cluster intact and resamples only the hub configuration, also reports the star as “hierarchical”: a near-regular simplex of hubs already minimizes  $\hat{\delta}$ , so any hub randomization raises it. Null B is therefore not a valid depth test as constructed and is not used in the paper; a star-calibrated version (excess B relative to a matched star with the same cluster count and tightness) may still serve as a depth test, since under the Haar construction the two-level hierarchy sits  $2\text{--}3\times$  below the star. That version is run on the real backbones in Table 9. Under the plain Gaussian null the mid-radius star scores  $-0.104$  and the  $6\times 5$  hierarchy  $-0.14$ : a star and a two-level tree differ by a fraction of either’s excess, which is why depth cannot be read from the spectrum excess alone.

| model    | CIFAR-100 ( $K=20$ coarse) |        |               | ImageNet ( $K=30$ WordNet) |        |               |
|----------|----------------------------|--------|---------------|----------------------------|--------|---------------|
|          | real                       | star   | depth ( $z$ ) | real                       | star   | depth ( $z$ ) |
| ViT-T    | -0.024                     | -0.030 | +0.006 (+0.4) | -0.007                     | -0.006 | -0.001 (-0.1) |
| ViT-S    | -0.007                     | -0.044 | +0.036 (+2.4) | -0.017                     | -0.008 | -0.009 (-1.0) |
| ViT-B    | +0.004                     | -0.056 | +0.060 (+4.5) | -0.024                     | -0.013 | -0.011 (-0.9) |
| ViT-L    | -0.016                     | -0.056 | +0.040 (+4.1) | -0.025                     | -0.013 | -0.012 (-1.3) |
| DINO-B   | -0.036                     | -0.043 | +0.007 (+0.9) | -0.017                     | -0.006 | -0.010 (-1.6) |
| DINOv2-S | -0.020                     | -0.027 | +0.008 (+0.7) | -0.004                     | -0.001 | -0.003 (-0.2) |
| DINOv2-B | +0.012                     | -0.027 | +0.039 (+5.2) | -0.010                     | -0.002 | -0.008 (-0.8) |
| DINOv2-L | +0.015                     | -0.027 | +0.042 (+7.0) | -0.033                     | -0.004 | -0.029 (-2.5) |
| DINOv2-G | -0.003                     | -0.030 | +0.027 (+3.9) | -0.026                     | -0.005 | -0.020 (-1.7) |
| CLIP-B   | -0.025                     | -0.044 | +0.019 (+1.7) | -0.014                     | -0.010 | -0.004 (-0.4) |
| CLIP-L   | -0.013                     | -0.049 | +0.036 (+7.2) | -0.012                     | -0.012 | -0.001 (-0.1) |
| SigLIP-B | -0.023                     | -0.050 | +0.027 (+2.9) | -0.009                     | -0.010 | +0.001 (+0.1) |

Table 9: Star-calibrated depth test (excess B, Haar construction). For each backbone the class centroids are compared with a *matched star*:  $K$  Gaussian hubs with the real hubs’ RMS radius and, around each hub, an isotropic Gaussian cloud with that superclass’s real within-cluster RMS spread (same  $n$ ,  $d$ ,  $K$  and cluster sizes; 3 star seeds). “real” and “star” are excesses of  $\hat{\delta}$  over the hub-randomizing null (each cluster kept intact, hubs replaced by a Haar-rotated sample with the hubs’ exact spectrum, 10 replicates), which has a small star bias of its own; “depth” is their difference (real minus star; negative = more tree-like than a matched star), with  $z$  against the combined star, null and estimator spread. On ImageNet only DINOv2-L exceeds its matched star beyond two spreads ( $z = -2.5$ ; DINOv2-G  $-1.7$ , all others  $|z| \leq 1.6$ ); on CIFAR-100 every backbone is *less* tree-like than its matched star. The excess of §4 is therefore accounted for by clustered, star-like organization; residual depth is at most marginal.

| model      | CIFAR-100 images ( $K=20$ ) |               |                           | ImageNet images ( $K=30$ ) |               |                           |
|------------|-----------------------------|---------------|---------------------------|----------------------------|---------------|---------------------------|
|            | exc. ( $z$ )                | depth ( $z$ ) | $\hat{\delta}$ nat./Eucl. | exc. ( $z$ )               | depth ( $z$ ) | $\hat{\delta}$ nat./Eucl. |
| MERU ViT-S | -0.040 (-3.0)               | +0.023 (+1.7) | 0.163/0.163               | -0.024 (-2.5)              | -0.004 (-0.4) | 0.128/0.128               |
| CLIP ViT-S | -0.059 (-4.9)               | -0.001 (-0.1) | 0.140/0.141               | -0.025 (-2.7)              | -0.005 (-0.5) | 0.151/0.126               |
| MERU ViT-B | -0.042 (-3.4)               | +0.023 (+2.5) | 0.156/0.156               | -0.022 (-2.4)              | +0.000 (+0.0) | 0.129/0.129               |
| CLIP ViT-B | -0.056 (-4.4)               | +0.010 (+0.9) | 0.146/0.141               | -0.025 (-2.3)              | -0.001 (-0.1) | 0.151/0.124               |
| MERU ViT-L | -0.023 (-1.5)               | +0.043 (+3.1) | 0.170/0.171               | -0.021 (-2.2)              | +0.004 (+0.4) | 0.128/0.128               |
| CLIP ViT-L | -0.037 (-2.9)               | +0.025 (+2.1) | 0.170/0.155               | -0.025 (-2.5)              | -0.003 (-0.3) | 0.150/0.123               |

Table 10: Hyperbolic-backbone control. MERU (Desai et al., 2023) embeds images on the Lorentz hyperboloid with an entailment objective; its released Euclidean twin (a CLIP baseline: same ViT, same RedCaps data and recipe, minus the hyperbolic lift and entailment loss) differs only in geometry. Both pass through the paper’s instrument unchanged (class centroids of the projected embeddings; “exc.” is the excess over the spectrum-matched null, “depth” the star-calibrated test of Table 9, “nat.” the Gromov  $\hat{\delta}$  computed with the model’s own metric: Lorentz distance between tangent-space-mean centroids for MERU, angular for CLIP). Training in hyperbolic space changes nothing at the class level: MERU shows the same clustering excess as its twin, no residual depth beyond a matched star on either dataset, and its Lorentz-metric  $\hat{\delta}$  is indistinguishable from the Euclidean one. MERU’s hierarchy is a generic  $\rightarrow$  specific (text  $\supset$  image) partial order, not a class taxonomy; the class-tree depth that no standard backbone shows is not present in a hyperbolic one either. The synthetic two-level tree of Table 8 remains the existence proof that the depth test fires when depth is present.

## A.6 TEXT CENSUS

| model          | type      | $d$  | $\delta_{\text{norm}}$ | null              | excess | $z$  |
|----------------|-----------|------|------------------------|-------------------|--------|------|
| GPT-2 S        | causal LM | 768  | $0.131 \pm 0.007$      | $0.083 \pm 0.002$ | +0.048 | +6.5 |
| GPT-2 M        | causal LM | 1024 | $0.091 \pm 0.004$      | $0.103 \pm 0.002$ | -0.013 | -2.7 |
| GPT-2 L        | causal LM | 1280 | $0.095 \pm 0.004$      | $0.129 \pm 0.006$ | -0.034 | -5.0 |
| GPT-2 XL       | causal LM | 1600 | $0.083 \pm 0.005$      | $0.131 \pm 0.007$ | -0.048 | -5.4 |
| Pythia-410M    | causal LM | 1024 | $0.098 \pm 0.005$      | $0.136 \pm 0.005$ | -0.037 | -5.4 |
| Pythia-1B      | causal LM | 2048 | $0.096 \pm 0.004$      | $0.131 \pm 0.004$ | -0.035 | -6.2 |
| Pythia-2.8B    | causal LM | 2560 | $0.098 \pm 0.004$      | $0.139 \pm 0.008$ | -0.041 | -4.6 |
| OLMo-1B        | causal LM | 2048 | $0.100 \pm 0.005$      | $0.101 \pm 0.002$ | -0.001 | -0.2 |
| OLMo-7B        | causal LM | 4096 | $0.093 \pm 0.005$      | $0.108 \pm 0.002$ | -0.015 | -2.8 |
| BGE-base       | embedder  | 768  | $0.121 \pm 0.006$      | $0.117 \pm 0.005$ | +0.005 | +0.6 |
| BGE-large      | embedder  | 1024 | $0.121 \pm 0.008$      | $0.120 \pm 0.003$ | +0.001 | +0.1 |
| GTE-base       | embedder  | 768  | $0.124 \pm 0.011$      | $0.113 \pm 0.003$ | +0.011 | +0.9 |
| GTE-large      | embedder  | 1024 | $0.120 \pm 0.012$      | $0.116 \pm 0.003$ | +0.004 | +0.4 |
| E5-base        | embedder  | 768  | $0.115 \pm 0.007$      | $0.112 \pm 0.005$ | +0.003 | +0.3 |
| E5-large       | embedder  | 1024 | $0.114 \pm 0.010$      | $0.113 \pm 0.003$ | +0.001 | +0.1 |
| GTE-Qwen2-1.5B | embedder  | 1536 | $0.107 \pm 0.004$      | $0.126 \pm 0.004$ | -0.019 | -3.1 |

Table 11: Text models under the calibrated protocol (1000 ImageNet class prompts):  $\delta_{\text{norm}}$ , spectrum-matched null (3 reps) and excess.

| model          | re-extraction, 20 replicates |        |      |            | original extraction, 3 replicates (Table 11) |        |                      |
|----------------|------------------------------|--------|------|------------|----------------------------------------------|--------|----------------------|
|                | $\hat{\delta}$               | excess | $z$  | reps above | $\hat{\delta}$                               | excess | $\Delta\hat{\delta}$ |
| GPT-2 S        | 0.128                        | -0.015 | -1.6 | 20/20      | 0.131                                        | +0.048 | -0.003               |
| GPT-2 M        | 0.082                        | -0.010 | -1.2 | 18/20      | 0.091                                        | -0.013 | -0.009               |
| GPT-2 L        | 0.082                        | -0.041 | -5.3 | 20/20      | 0.095                                        | -0.034 | -0.013               |
| GPT-2 XL       | 0.072                        | -0.049 | -7.4 | 20/20      | 0.083                                        | -0.048 | -0.011               |
| Pythia-410M    | 0.098                        | -0.036 | -4.5 | 20/20      | 0.098                                        | -0.037 | +0.000               |
| Pythia-1B      | 0.096                        | -0.032 | -4.4 | 20/20      | 0.096                                        | -0.035 | +0.000               |
| Pythia-2.8B    | 0.098                        | -0.036 | -4.3 | 20/20      | 0.098                                        | -0.041 | -0.000               |
| OLMo-1B        | 0.082                        | -0.039 | -5.7 | 20/20      | 0.100                                        | -0.001 | -0.018               |
| OLMo-7B        | -                            | -      | -    | -          | 0.093                                        | -0.015 | -                    |
| BGE-base       | 0.121                        | +0.004 | +0.5 | 5/20       | 0.121                                        | +0.005 | +0.000               |
| BGE-large      | 0.121                        | +0.003 | +0.3 | 5/20       | 0.121                                        | +0.001 | -0.000               |
| GTE-base       | 0.124                        | +0.011 | +1.0 | 0/20       | 0.124                                        | +0.011 | +0.000               |
| GTE-large      | 0.120                        | +0.007 | +0.6 | 1/20       | 0.120                                        | +0.004 | -0.000               |
| GTE-Qwen2-1.5B | 0.122                        | +0.002 | +0.2 | 8/20       | 0.107                                        | -0.019 | +0.015               |
| E5-base        | 0.115                        | +0.002 | +0.3 | 9/20       | 0.115                                        | +0.003 | -0.000               |
| E5-large       | 0.114                        | +0.005 | +0.4 | 3/20       | 0.114                                        | +0.001 | +0.000               |

Table 12: The text census independently re-extracted (same template and pooling; padding-free (batch-size-1) inference) and re-scored with 20 spectrum-null replicates; OLMo-7B not re-run (exceeds the local GPU). Raw  $\hat{\delta}$  differs from the original extraction because the embeddings were re-extracted, not because of the replicate count;  $\Delta\hat{\delta}$  is that extraction-to-extraction difference. Sign agreement with the 3-replicate census: 13/15. Large  $|z|$  read as “below every replicate”. Three verdicts change between extractions, each with a known cause: GPT-2 S (above null in the left-padded original, at null padding-free; Table 13), OLMo-1B (the original used a different checkpoint revision; both new extractions agree at -0.039), and GTE-Qwen2 (genuine in the original with 3 replicates, -0.019; at null in both re-extractions, +0.002 with rank 8/20 here; not used as evidence).

| batch size     | GPT-2 M             |                     | OLMo-1B             |                     |
|----------------|---------------------|---------------------|---------------------|---------------------|
|                | fp32 $\hat{\delta}$ | fp16 $\hat{\delta}$ | fp32 $\hat{\delta}$ | fp16 $\hat{\delta}$ |
| 1 (no padding) | 0.0816              | 0.0817              | 0.0817              | 0.0817              |
| 16             | 0.0996              | 0.0997              | 0.0817              | 0.0816              |
| 32             | 0.0905              | 0.0905              | 0.0817              | 0.0817              |

Table 13: Where extraction-to-extraction variability comes from. Same 1000 prompts, same model; only batch size and precision vary. Precision is irrelevant (fp16 equals fp32 to four decimals). For GPT-2, left-padded batching changes the hidden states (mean cosine to the padding-free extraction 0.98) and moves  $\hat{\delta}$  by up to 0.018; OLMo handles left padding correctly and is invariant. Padding-free (batch-size-1) extraction is therefore the protocol of Table 12.

#### A.7 CLASS-COUNT VS HIERARCHY DEPTH

| model    | $C$  | random subsets (span the hierarchy) |        |        | WordNet-coherent (siblings) |        |        |
|----------|------|-------------------------------------|--------|--------|-----------------------------|--------|--------|
|          |      | $\delta_{\text{norm}}$              | excess | NC adv | $\delta_{\text{norm}}$      | excess | NC adv |
| DINOv2-L | 10   | 0.030                               | -0.121 | +1.04  | 0.068                       | -0.096 | +0.08  |
| DINOv2-L | 20   | 0.040                               | -0.157 | +1.32  | 0.080                       | -0.118 | +0.12  |
| DINOv2-L | 50   | 0.049                               | -0.124 | +1.58  | 0.087                       | -0.093 | +0.53  |
| DINOv2-L | 100  | 0.060                               | -0.081 | +1.46  | 0.085                       | -0.061 | +0.77  |
| DINOv2-L | 200  | 0.062                               | -0.054 | +1.45  | 0.087                       | -0.034 | +0.59  |
| DINOv2-L | 500  | 0.061                               | -0.026 | +1.01  | 0.070                       | -0.021 | +0.82  |
| DINOv2-L | 1000 | 0.059                               | -0.016 | +0.57  | —                           | —      | —      |
| DINOv2-G | 10   | 0.029                               | -0.123 | +0.80  | 0.072                       | -0.094 | -0.56  |
| DINOv2-G | 20   | 0.042                               | -0.155 | +1.30  | 0.077                       | -0.120 | -0.18  |
| DINOv2-G | 50   | 0.060                               | -0.111 | +1.51  | 0.090                       | -0.092 | +0.26  |
| DINOv2-G | 100  | 0.068                               | -0.073 | +1.30  | 0.090                       | -0.056 | +0.36  |
| DINOv2-G | 200  | 0.076                               | -0.039 | +1.28  | 0.090                       | -0.033 | +0.27  |
| DINOv2-G | 500  | 0.082                               | -0.003 | +0.49  | 0.077                       | -0.013 | +0.22  |
| DINOv2-G | 1000 | 0.076                               | +0.003 | -0.13  | —                           | —      | —      |
| CLIP-L   | 10   | 0.071                               | -0.093 | -0.08  | 0.099                       | -0.097 | -0.36  |
| CLIP-L   | 20   | 0.090                               | -0.099 | +0.20  | 0.129                       | -0.067 | -0.24  |
| CLIP-L   | 50   | 0.110                               | -0.076 | +0.34  | 0.147                       | -0.051 | -0.46  |
| CLIP-L   | 100  | 0.115                               | -0.051 | +0.21  | 0.130                       | -0.040 | -0.31  |
| CLIP-L   | 200  | 0.113                               | -0.038 | +0.00  | 0.124                       | -0.025 | -0.35  |
| CLIP-L   | 500  | 0.113                               | -0.031 | -0.27  | 0.114                       | -0.027 | -0.42  |
| CLIP-L   | 1000 | 0.112                               | -0.023 | -0.53  | —                           | —      | —      |

Table 14: Class-count vs hierarchy-depth deconfound on ImageNet (advantages in pp; means over subset seeds;  $C \leq 30$  uses exact quadruple enumeration).

#### A.8 TREE-MAP METRIC AND LINKAGE CONTROLS

Threshold sensitivity of the selection criterion of §5: the selected configuration is stable for degeneracy bounds in  $[0.45, 0.50]$  on ImageNet and  $[0.40, 0.50]$  on CIFAR-100; tighter bounds select cosine-complete on ImageNet (gap 0.13) and cosine-Ward on CIFAR-100 (gap 0.56), and in every case the label-free gap is moderate and configuration-sensitive, never a categorical island. The criterion must be applied per dataset: the configuration it selects on DBpedia (Euclidean, average linkage, where nothing is degenerate) is exactly the degenerate naive configuration on ImageNet, so a criterion transferred across datasets would reintroduce the island.

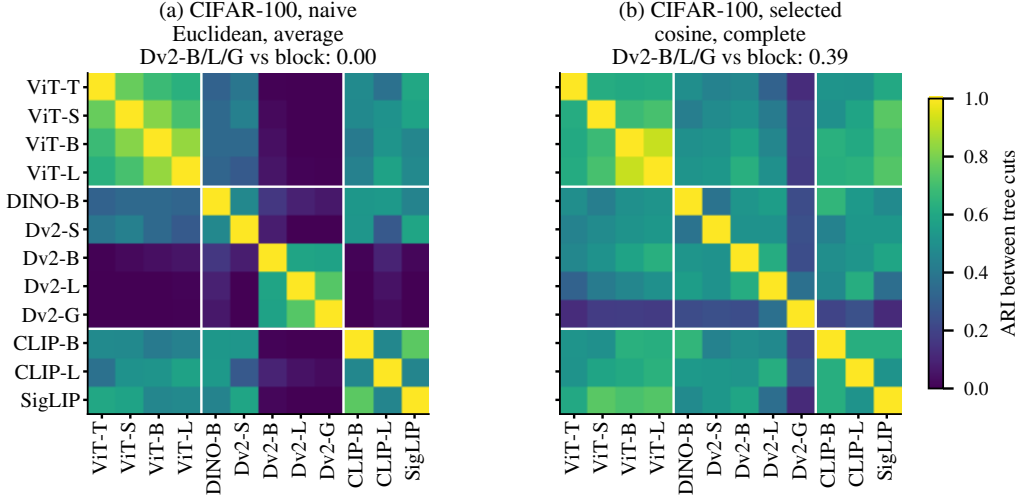


Figure 7: **The island is an artifact on CIFAR-100 too.** Same reading as Figure 6 on the 20 CIFAR-100 superclass cuts: naive (a) vs criterion-selected (b).

| dataset  | configuration              | max frac. | CPCC  | Dv2-B/L/G vs block | Dv2-S+DINO-B vs block | within block |
|----------|----------------------------|-----------|-------|--------------------|-----------------------|--------------|
| imagenet | euclid-average (deg.)      | 0.97      | 0.761 | 0.03               | 0.08                  | 0.64         |
| imagenet | euclid-complete (deg.)     | 0.83      | 0.606 | 0.03               | 0.17                  | 0.44         |
| imagenet | euclid-ward                | 0.48      | 0.464 | 0.15               | 0.38                  | 0.53         |
| imagenet | cosine-average (selected)  | 0.40      | 0.691 | 0.38               | 0.35                  | 0.48         |
| imagenet | cosine-complete            | 0.32      | 0.539 | 0.13               | 0.17                  | 0.50         |
| imagenet | cosine-ward                | 0.37      | 0.464 | 0.17               | 0.39                  | 0.55         |
| cifar100 | euclid-average (deg.)      | 0.80      | 0.798 | 0.00               | 0.41                  | 0.57         |
| cifar100 | euclid-complete (deg.)     | 0.66      | 0.722 | 0.17               | 0.40                  | 0.54         |
| cifar100 | euclid-ward                | 0.33      | 0.534 | 0.37               | 0.65                  | 0.63         |
| cifar100 | cosine-average (deg.)      | 0.53      | 0.777 | 0.30               | 0.54                  | 0.58         |
| cifar100 | cosine-complete (selected) | 0.39      | 0.698 | 0.39               | 0.51                  | 0.63         |
| cifar100 | cosine-ward                | 0.14      | 0.584 | 0.56               | 0.62                  | 0.61         |

Table 15: Tree-map configurations with the two quantities that drive the selection criterion of §5: the degeneracy diagnostic (largest-cluster fraction at the reference cut, worst model) and the mean cophenetic fidelity (CPCC) of each model’s dendrogram to its own distance matrix. Configurations with max-cluster fraction  $> 0.5$  are excluded; among the rest, highest CPCC selects the marked row. Mean pairwise ARIs at the reference cut shown for all configurations.

| model    | euclid-avg (naive, degen.) | euclid-ward | cosine-avg (degen. C100) | cosine-complete $\leftarrow$ sel. | cosine-ward |
|----------|----------------------------|-------------|--------------------------|-----------------------------------|-------------|
| ViT-T    | 0.31                       | 0.38        | 0.29                     | 0.33                              | 0.33        |
| ViT-S    | 0.35                       | 0.56        | 0.37                     | 0.47                              | 0.56        |
| ViT-B    | 0.39                       | 0.61        | 0.43                     | 0.45                              | 0.57        |
| ViT-L    | 0.38                       | 0.55        | 0.49                     | 0.48                              | 0.60        |
| DINO-B   | 0.22                       | 0.48        | 0.28                     | 0.38                              | 0.54        |
| DINOv2-S | 0.12                       | 0.51        | 0.32                     | 0.33                              | 0.49        |
| DINOv2-B | 0.02                       | 0.47        | 0.35                     | 0.42                              | 0.51        |
| DINOv2-L | 0.01                       | 0.32        | 0.25                     | 0.50                              | 0.48        |
| DINOv2-G | 0.00                       | 0.19        | 0.08                     | 0.20                              | 0.52        |
| CLIP-B   | 0.18                       | 0.48        | 0.18                     | 0.37                              | 0.48        |
| CLIP-L   | 0.35                       | 0.52        | 0.32                     | 0.41                              | 0.54        |
| SigLIP-B | 0.22                       | 0.49        | 0.22                     | 0.37                              | 0.52        |

Table 16: Recovery of the true CIFAR-100 superclasses (ARI of the 20-cluster cut) per model and configuration, the quantity behind the corrected picture of §5. Degenerate configurations marked; the criterion-selected configuration is cosine-complete.

## A.9 TASK ACCURACIES (REGENERATED)

| model    | IN        | C100      | C10       | DTD       | FMNIST    | MNIST     |
|----------|-----------|-----------|-----------|-----------|-----------|-----------|
| ViT-T    | 68.1/67.2 | 29.3/28.2 | 87.5/87.4 | 63.7/62.5 | 78.5/78.4 | 75.9/74.0 |
| ViT-S    | 77.7/77.5 | 74.9/74.6 | 93.4/93.2 | 69.3/68.5 | 80.7/80.4 | 84.8/84.1 |
| ViT-B    | 81.6/81.7 | 77.3/77.0 | 94.5/94.0 | 70.8/70.8 | 81.3/81.6 | 81.6/80.5 |
| ViT-L    | 86.1/86.2 | 85.3/85.2 | 97.5/97.5 | 71.8/71.5 | 83.7/83.5 | 84.2/83.6 |
| DINOv2-S | 74.7/74.4 | 73.4/74.4 | 95.0/95.2 | 73.4/73.4 | 82.2/82.2 | 83.4/81.9 |
| DINOv2-B | 78.9/78.9 | 87.4/87.9 | 97.1/97.4 | 74.2/74.7 | 83.2/82.8 | 80.6/78.8 |
| DINOv2-L | 79.8/80.4 | 85.6/87.7 | 97.9/98.3 | 74.4/75.2 | 84.4/84.5 | 74.1/72.7 |
| DINOv2-G | 78.4/78.3 | 89.8/91.0 | 97.8/98.2 | 72.9/73.1 | 85.9/85.8 | 79.3/78.6 |
| CLIP-B   | 61.4/59.9 | 65.6/65.1 | 87.9/88.3 | 63.7/63.2 | 75.9/75.1 | 70.1/66.6 |
| CLIP-L   | 75.0/74.5 | 74.6/74.5 | 95.9/95.9 | 68.6/69.0 | 76.2/75.9 | 85.2/86.1 |

Table 17: Nearest-centroid accuracy (%), Euclidean/Poincaré, regenerated from the audited reruns.

| model    | IN        | C100      | C10       | DTD       | FMNIST    | MNIST     |
|----------|-----------|-----------|-----------|-----------|-----------|-----------|
| ViT-T    | 97.5/97.6 | 62.4/63.1 | 87.6/88.1 | 83.3/83.9 | 81.6/81.9 | 74.3/74.6 |
| ViT-S    | 98.6/98.7 | 92.2/92.4 | 92.0/92.5 | 85.4/86.0 | 83.7/83.5 | 80.5/81.4 |
| ViT-B    | 98.9/99.0 | 92.5/92.7 | 92.7/93.0 | 85.4/86.2 | 83.4/83.3 | 76.8/77.6 |
| ViT-L    | 99.4/99.4 | 94.8/95.5 | 95.0/95.8 | 85.6/86.5 | 84.4/84.7 | 80.8/81.6 |
| DINOv2-S | 97.4/97.9 | 94.3/95.1 | 92.9/93.5 | 89.8/90.5 | 85.4/85.5 | 80.2/80.8 |
| DINOv2-B | 97.0/97.8 | 96.5/97.2 | 94.7/95.4 | 89.2/90.3 | 85.9/85.9 | 76.6/77.1 |
| DINOv2-L | 96.2/97.1 | 96.1/97.1 | 94.1/95.1 | 87.1/88.4 | 87.8/87.9 | 70.5/71.1 |
| DINOv2-G | 96.2/97.1 | 95.9/97.2 | 92.3/93.8 | 87.2/88.5 | 89.0/89.2 | 72.9/73.8 |
| CLIP-B   | 96.8/96.8 | 87.7/88.9 | 89.5/90.1 | 82.2/83.1 | 78.8/78.9 | 70.9/69.7 |
| CLIP-L   | 98.4/98.6 | 90.0/90.8 | 92.4/93.0 | 85.2/86.2 | 76.8/77.3 | 83.7/84.7 |

Table 18: 5-way 5-shot accuracy (%), Euclidean/Poincaré, 1000 paired episodes.

## A.10 METRIC CONTROLS

| model    | IN    | C100  | C10   | DTD   | FMNIST | MNIST |
|----------|-------|-------|-------|-------|--------|-------|
| ViT-T    | -0.01 | +1.01 | +0.66 | -0.23 | +0.21  | +0.64 |
| ViT-S    | +0.17 | +0.59 | +0.56 | -0.03 | +0.28  | +0.34 |
| ViT-B    | +0.20 | +0.56 | +0.49 | +0.28 | +0.03  | +0.19 |
| ViT-L    | +0.11 | +1.17 | +1.25 | +0.13 | +0.00  | +0.08 |
| DINOv2-S | +0.68 | +0.80 | -0.36 | +0.30 | +0.02  | -0.07 |
| DINOv2-B | +1.40 | +1.11 | +0.27 | +1.48 | +0.21  | +0.17 |
| DINOv2-L | +2.13 | +1.80 | +0.92 | +2.40 | +0.29  | +0.25 |
| DINOv2-G | +2.57 | +2.44 | +2.51 | +2.63 | +0.38  | +0.30 |
| CLIP-B   | -0.04 | -0.03 | -0.19 | -0.23 | +0.06  | +0.04 |
| CLIP-L   | +0.01 | +0.12 | +0.16 | -0.08 | +0.12  | +0.08 |

Table 19: Few-shot advantage of cosine over Euclidean (pp).

| model    | IN    | C100  | C10   | DTD   | FMNIST | MNIST |
|----------|-------|-------|-------|-------|--------|-------|
| ViT-T    | +0.01 | +0.45 | +0.19 | +0.32 | +0.10  | +0.52 |
| ViT-S    | -0.02 | +0.16 | +0.27 | +0.25 | +0.08  | +0.65 |
| ViT-B    | -0.04 | +0.12 | +0.18 | +0.19 | +0.09  | +0.51 |
| ViT-L    | -0.03 | +0.31 | +0.35 | +0.16 | +0.17  | +0.42 |
| DINOv2-S | +0.03 | +0.15 | +0.13 | +0.04 | +0.07  | +0.45 |
| DINOv2-B | +0.04 | +0.09 | +0.03 | +0.01 | +0.04  | +0.48 |
| DINOv2-L | +0.02 | +0.09 | -0.00 | +0.03 | +0.03  | +0.43 |
| DINOv2-G | +0.03 | +0.13 | -0.02 | +0.01 | +0.04  | +0.56 |
| CLIP-B   | -0.00 | +0.44 | +0.20 | +0.36 | +0.17  | +0.23 |
| CLIP-L   | +0.00 | +0.39 | +0.25 | +0.36 | +0.33  | +0.58 |

Table 20: Few-shot advantage of the radial-rescaling control over Euclidean (pp): the map without the metric changes nothing.

#### A.11 STATISTICS

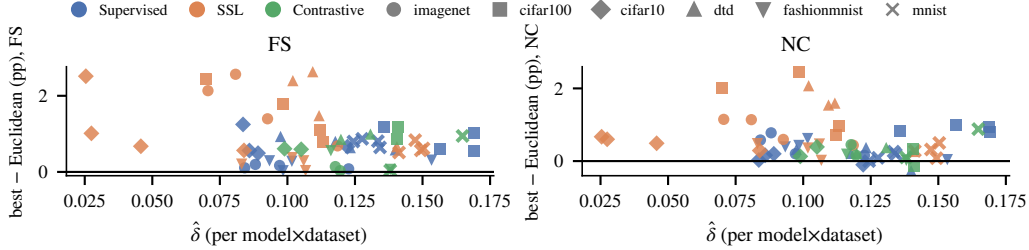


Figure 8: **More tree-like cells carry larger zero-cost metric gains.** Best zero-cost metric minus Euclidean per cell vs  $\delta_{\text{norm}}$  (FS left, NC right). Within-dataset correlations  $-0.60$  to  $-0.83$  (FS) and  $-0.61$  to  $-0.86$  (NC); pooled correlations are family-driven and not used as evidence (Appendix A.11).

| measure                          | raw   | null mean | $\tau_{0.05}$ | calibrated | null-centered | pairs $p < 0.05$ |
|----------------------------------|-------|-----------|---------------|------------|---------------|------------------|
| mutual-kNN ( $k=10$ ), Euclidean | 0.472 | 0.010     | 0.015         | 0.464      | 0.462         | 66/66            |
| mutual-kNN, Poincaré             | 0.425 | 0.010     | 0.019         | 0.414      | 0.415         | 66/66            |
| linear CKA, Euclidean            | 0.633 | 0.108     | 0.111         | 0.577      | 0.524         | 66/66            |
| linear CKA, Poincaré             | 0.640 | 0.108     | 0.111         | 0.586      | 0.532         | 66/66            |

Table 21: **Cross-model agreement survives permutation calibration.** Means over the 66 model pairs of the 12 vision backbones on the 1000 ImageNet class centroids (census cache). Calibration of Gröger et al. (2026) with  $K=200$  permutations of the class correspondence and  $\alpha=0.05$ , scalar (no layer search):  $\tau_{0.05}$  is the  $\lceil 0.95(K+1) \rceil$ -th order statistic of the observed score and its nulls (eq. 9),  $p = (1 + \#\{\text{null} \geq \text{obs}\}) / (K+1)$  (eq. 10), calibrated =  $\max\{(\text{obs} - \tau_{0.05}) / (1 - \tau_{0.05}), 0\}$  (eq. 12); “null-centered” is the earlier raw-minus-null-mean variant, kept for continuity. The mKNN null mean equals the analytic chance level  $k/(n-1) = 10/999$ .

| quantity  | dataset   | pooled $r$ | demeaned $r$ | ViT   | DINOv2 | CLIP |
|-----------|-----------|------------|--------------|-------|--------|------|
| NC H-R    | ImageNet  | -0.83      | -0.73        | -0.99 | -0.77  | (-1) |
|           | CIFAR-100 | -0.87      | -0.45        | -0.68 | -0.33  | (-1) |
|           | CIFAR-10  | -0.63      | -0.31        | +0.01 | -0.98  | (+1) |
|           | DTD       | -0.46      | -0.57        | -0.78 | -0.76  | (+1) |
| FS H-R    | ImageNet  | -0.57      | -0.73        | -0.04 | -0.97  | (-1) |
|           | CIFAR-100 | -0.67      | -0.63        | -0.41 | -0.96  | (+1) |
|           | CIFAR-10  | -0.78      | -0.64        | -0.22 | -0.79  | (-1) |
|           | DTD       | -0.71      | -0.86        | -0.95 | -0.98  | (+1) |
| FS best-R | ImageNet  | -0.64      | -0.71        | -0.57 | -0.90  | (-1) |
|           | CIFAR-100 | -0.83      | -0.82        | -0.58 | -0.96  | (+1) |
|           | CIFAR-10  | -0.63      | -0.55        | -0.26 | -0.65  | (-1) |
|           | DTD       | -0.60      | -0.70        | -0.95 | -0.88  | (+1) |

Table 22: Family-identity control for the  $\delta_{\text{norm}}$ -gain correlations (within dataset, 10 backbones = 3 families with nested scales). Demeaned: both variables centered within family before correlating; ViT/DINOv2: within-family correlations (4 models each); CLIP has only two models, so its within-family value (parenthesized sign) is not informative. Without DINOv2 (six backbones) the NC correlation holds on ImageNet and CIFAR-100 ( $-0.87/-0.65$ ) but not on CIFAR-10/DTD ( $+0.23/-0.34$ ). The within-dataset prediction survives the family control; pooled cross-dataset correlations of the best-metric advantage do not ( $-0.31 \rightarrow -0.18$  over all 60 cells,  $-0.33 \rightarrow -0.04$  on hierarchical cells) and are therefore family-driven.

| model    | IN             | C100           | C10            | DTD      | FMNIST        | MNIST          |
|----------|----------------|----------------|----------------|----------|---------------|----------------|
| ViT-T    | $10^{-22}$ (-) | $10^{-5}$ (-)  | 0.19 (-)       | 0.03 (-) | 0.38 (-)      | $10^{-13}$ (-) |
| ViT-S    | 0.09 (-)       | 0.10 (-)       | 0.02 (-)       | 0.10 (-) | 0.08 (-)      | $10^{-4}$ (-)  |
| ViT-B    | 0.21 (+)       | 0.07 (-)       | $10^{-5}$ (-)  | 1.00 (-) | 0.08 (+)      | $10^{-7}$ (-)  |
| ViT-L    | 0.58 (+)       | 0.59 (-)       | 0.76 (+)       | 0.69 (-) | 0.12 (-)      | 0.00 (-)       |
| DINOv2-S | 0.00 (-)       | $10^{-6}$ (+)  | 0.14 (+)       | 1.00 (+) | 1.00 (-)      | $10^{-11}$ (-) |
| DINOv2-B | 0.57 (+)       | $10^{-4}$ (+)  | $10^{-4}$ (+)  | 0.31 (+) | 0.01 (-)      | $10^{-16}$ (-) |
| DINOv2-L | $10^{-14}$ (+) | $10^{-38}$ (+) | $10^{-8}$ (+)  | 0.11 (+) | 0.40 (+)      | $10^{-12}$ (-) |
| DINOv2-G | 0.10 (-)       | $10^{-16}$ (+) | $10^{-13}$ (+) | 0.81 (+) | 0.72 (-)      | 0.00 (-)       |
| CLIP-B   | $10^{-40}$ (-) | 0.07 (-)       | $10^{-4}$ (+)  | 0.46 (-) | $10^{-4}$ (-) | $10^{-45}$ (-) |
| CLIP-L   | $10^{-8}$ (-)  | 0.66 (-)       | 0.59 (-)       | 0.45 (+) | 0.09 (-)      | $10^{-7}$ (+)  |

Table 23: McNemar  $p$ -values for NC test-set decisions, Poincaré vs Euclidean (order of magnitude when  $p < 10^{-3}$ ; sign of the NC advantage in parentheses, so significance is readable with its direction).

| task | dataset            | $r$   | CI95           |
|------|--------------------|-------|----------------|
| NC   | imagenet           | -0.83 | [-0.97, -0.65] |
| NC   | cifar100           | -0.87 | [-0.97, -0.78] |
| NC   | cifar10            | -0.63 | [-0.97, -0.06] |
| NC   | dtd                | -0.46 | [-0.90, +0.22] |
| FS   | imagenet           | -0.57 | [-0.91, +0.14] |
| FS   | cifar100           | -0.67 | [-0.95, -0.15] |
| FS   | cifar10            | -0.78 | [-0.93, -0.23] |
| FS   | dtd                | -0.71 | [-0.94, -0.32] |
| NC   | POOLED(cluster-bs) | -0.45 | [-0.60, -0.36] |
| FS   | POOLED(cluster-bs) | -0.18 | [-0.50, +0.12] |

Table 24: Bootstrap CI95 for the correlations between  $\delta_{\text{norm}}$  and the projection advantage (within-dataset: 10k resamples of the 10 models; pooled: cluster bootstrap over datasets).

| policy (test advantage over Euclidean, pp) | all 60 cells | hierarchical (40) |
|--------------------------------------------|--------------|-------------------|
| always cosine                              | +0.23        | +0.28             |
| objective rule (zero validation)           | +0.28        | +0.41             |
| $\delta$ -gated objective rule             | +0.08        | +0.13             |
| validation-picked per cell                 | +0.63        | +0.78             |
| test oracle (reference)                    | +0.64        | +0.78             |

Table 25: Metric-selection policies evaluated on held-out episodes (500 validation / 500 test per cell, paper protocol). Validation picking retains the oracle advantage; the zero-validation objective rule modestly beats always-cosine.

| NC policy (advantage over Euclidean, pp) | all 60 | flat (20) | hierarchical (40) |
|------------------------------------------|--------|-----------|-------------------|
| always cosine                            | +0.48  | +0.24     | +0.59             |
| always Poincaré                          | -0.24  | -0.70     | -0.00             |
| gated rule (flat→R)                      | +0.34  | +0.00     | +0.51             |
| $\delta$ -gated rule                     | +0.19  | +0.00     | +0.29             |

Table 26: NC metric-selection policies. The gate’s value is avoiding the Poincaré projection on flat-label data; cosine is a safe NC default everywhere.

#### A.12 PROJECTION-TARGET SENSITIVITY

| target $t$        | 0.25  | 0.35  | 0.41  | 0.45  | 0.50  | 0.58  | 0.71  | 0.95  |
|-------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| FS adv. mean (pp) | +0.88 | +0.87 | +0.87 | +0.86 | +0.84 | +0.82 | +0.76 | +0.56 |
| NC adv. mean (pp) | -0.30 | -0.25 | -0.23 | -0.21 | -0.17 | -0.12 | -0.01 | +0.15 |
| % cells FS > 0    | 97    | 100   | 100   | 100   | 100   | 100   | 100   | 100   |

Table 27: Sensitivity to the projection target  $t$  (36 hierarchical cells). The headline  $t = 1/\sqrt{2} \approx 0.71$  was fixed a priori and is conservative for FS.

## A.13 ESTIMATOR ROBUSTNESS

| model                                                                         | data | excess vs quadruple budget ( $10^4 \dots 10^6$ ) |         |                |        |                                  |        | p99.9 @ $5 \times 10^5$ |
|-------------------------------------------------------------------------------|------|--------------------------------------------------|---------|----------------|--------|----------------------------------|--------|-------------------------|
| ViT-L                                                                         | IN   | -0.026                                           | -0.024  | -0.027         | -0.030 | -0.034                           | -0.035 | -0.018                  |
| DINOv2-L                                                                      | IN   | -0.027                                           | -0.019  | -0.024         | -0.011 | -0.004                           | -0.005 | -0.024                  |
| CLIP-B                                                                        | IN   | -0.008                                           | -0.012  | -0.023         | -0.025 | -0.025                           | -0.031 | -0.007                  |
| ViT-L                                                                         | C100 | -0.045                                           | -0.047  | -0.047         | -0.049 | -0.052                           | -0.053 | -0.033                  |
| DINOv2-L                                                                      | C100 | -0.071                                           | -0.055  | -0.054         | -0.050 | -0.052                           | -0.055 | -0.069                  |
| CLIP-B                                                                        | C100 | -0.045                                           | -0.047  | -0.050         | -0.049 | -0.052                           | -0.051 | -0.030                  |
| ViT-L                                                                         | DTD  | -0.064                                           | -0.070  | -0.076         | -0.083 | -0.084                           | -0.084 | -0.054                  |
| DINOv2-L                                                                      | DTD  | -0.068                                           | -0.072  | -0.075         | -0.083 | -0.084                           | -0.084 | -0.064                  |
| CLIP-B                                                                        | DTD  | -0.048                                           | -0.055  | -0.059         | -0.065 | -0.066                           | -0.066 | -0.035                  |
| Centroid bootstrap (30 resamples of the per-class images): s.d. of the excess |      |                                                  |         |                |        |                                  |        |                         |
| ViT-L                                                                         | C100 | excess                                           | -0.0493 | bootstrap s.d. | 0.0029 | (30 resamples, 500 images/class) |        |                         |
| DINOv2-L                                                                      | C100 | excess                                           | -0.0493 | bootstrap s.d. | 0.0022 | (30 resamples, 500 images/class) |        |                         |
| DINOv2-G                                                                      | C100 | excess                                           | -0.0778 | bootstrap s.d. | 0.0020 | (30 resamples, 500 images/class) |        |                         |
| CLIP-B                                                                        | C100 | excess                                           | -0.0476 | bootstrap s.d. | 0.0017 | (30 resamples, 500 images/class) |        |                         |
| ViT-L                                                                         | DTD  | excess                                           | -0.0803 | bootstrap s.d. | 0.0049 | (30 resamples, 80 images/class)  |        |                         |
| DINOv2-L                                                                      | DTD  | excess                                           | -0.0878 | bootstrap s.d. | 0.0048 | (30 resamples, 80 images/class)  |        |                         |
| DINOv2-G                                                                      | DTD  | excess                                           | -0.0863 | bootstrap s.d. | 0.0051 | (30 resamples, 80 images/class)  |        |                         |
| CLIP-B                                                                        | DTD  | excess                                           | -0.0581 | bootstrap s.d. | 0.0054 | (30 resamples, 80 images/class)  |        |                         |

Table 28: Estimator robustness. Top: on CIFAR-100 and DTD the excess is stable once the quadruple budget reaches  $10^5$  (the paper uses  $5 \times 10^5$ ); on ImageNet ( $C=1000$ ) every sign holds at every budget but magnitudes drift by up to 0.02 (DINOv2-L toward its null, ViT-L and CLIP-B away from it), so ImageNet excess magnitudes carry a budget caveat. The 99.9th-percentile variant of the four-point statistic, robust to the supremum, is negative wherever the supremum excess is. Bottom: resampling the per-class images that form each centroid (reported on CIFAR-100 and DTD) moves the excess by an order of magnitude less than its size.

| model    | p99.9 $\hat{\delta}$ | excess  | $z$   | null reps above real |
|----------|----------------------|---------|-------|----------------------|
| ViT-T    | 0.100                | +0.0069 | +1.3  | 1/20                 |
| ViT-S    | 0.079                | -0.0048 | -1.1  | 17/20                |
| ViT-B    | 0.081                | -0.0065 | -1.8  | 20/20                |
| ViT-L    | 0.069                | -0.0139 | -4.3  | 20/20                |
| DINO-B   | 0.054                | -0.0247 | -10.9 | 20/20                |
| DINOv2-S | 0.053                | -0.0220 | -16.1 | 20/20                |
| DINOv2-B | 0.035                | -0.0256 | -33.1 | 20/20                |
| DINOv2-L | 0.032                | -0.0235 | -42.2 | 20/20                |
| DINOv2-G | 0.032                | -0.0203 | -37.9 | 20/20                |
| CLIP-B   | 0.087                | -0.0061 | -1.6  | 19/20                |
| CLIP-L   | 0.076                | -0.0114 | -3.1  | 20/20                |
| SigLIP-B | 0.076                | -0.0090 | -2.7  | 20/20                |

Table 29: The ImageNet census under the supremum-robust statistic: 99.9th percentile of the four-point defect, 20 spectrum-null replicates, with the percentile rank (how many replicates exceed the real value; 20/20 = below every replicate). The ordering shifts relative to the supremum: DINO/DINOv2 carry the most budget-robust excess, while ViT-T’s supremum excess does not survive (+0.007, above 19/20 replicates), consistent with the ordering of the norm-structure descriptor  $\xi$  (Table 39). Large  $|z|$  should be read as “below every replicate”, not as a Gaussian tail probability.

|          | IN     |    | C100   |    | C10    |    | DTD    |    | FMNIST |    | MNIST  |    |
|----------|--------|----|--------|----|--------|----|--------|----|--------|----|--------|----|
| model    | exc.   | r  | exc.   | r  | exc.   | r  | exc.   | r  | exc.   | r  | exc.   | r  |
| ViT-T    | -0.019 | 20 | -0.039 | 20 | -0.051 | 20 | -0.042 | 20 | -0.012 | 11 | -0.041 | 19 |
| ViT-S    | -0.021 | 20 | -0.031 | 20 | -0.090 | 20 | -0.055 | 20 | -0.060 | 20 | -0.051 | 20 |
| ViT-B    | -0.027 | 20 | -0.015 | 18 | -0.083 | 20 | -0.061 | 20 | -0.069 | 20 | -0.041 | 20 |
| ViT-L    | -0.031 | 20 | -0.043 | 20 | -0.096 | 20 | -0.082 | 20 | -0.079 | 20 | -0.060 | 20 |
| DINO-B   | -0.005 | 16 | -0.063 | 20 | -0.094 | 20 | -0.057 | 20 | -0.035 | 18 | -0.058 | 20 |
| DINOv2-S | +0.012 | 0  | -0.051 | 20 | -0.095 | 20 | -0.046 | 20 | -0.061 | 20 | -0.029 | 17 |
| DINOv2-B | +0.007 | 0  | -0.042 | 20 | -0.134 | 20 | -0.071 | 20 | -0.089 | 20 | -0.036 | 18 |
| DINOv2-L | -0.008 | 20 | -0.051 | 20 | -0.154 | 20 | -0.081 | 20 | -0.068 | 20 | -0.032 | 19 |
| DINOv2-G | +0.004 | 0  | -0.079 | 20 | -0.155 | 20 | -0.073 | 20 | -0.075 | 20 | -0.037 | 19 |
| CLIP-B   | -0.027 | 20 | -0.044 | 20 | -0.071 | 20 | -0.062 | 20 | -0.042 | 19 | -0.028 | 16 |
| CLIP-L   | -0.020 | 20 | -0.038 | 20 | -0.080 | 20 | -0.056 | 20 | -0.058 | 20 | -0.023 | 15 |
| SigLIP-B | -0.017 | 20 | -0.037 | 20 | -0.088 | 20 | -0.062 | 20 | -0.068 | 20 | -0.067 | 20 |

Table 30: **The 20-replicate census: excess and percentile rank per cell.** Excess over the spectrum-matched null and rank r (number of the 20 null replicates above the real value; 20 = below every replicate). 57/72 cells lie below every replicate. Secondary  $z$  summary (excess over the combined null and estimator s.d.):  $|z| \geq 2$  in 50/72 cells,  $z \leq -3$  in 31/48 hierarchical-dataset cells. Agreement with the 5-replicate census (Table 6) on the 72 comparable cells: sign 72/72, significance at  $|z| \geq 2$  67/72. Every cell is computed on the census cache (one centroid source; the ImageNet cache was regenerated with the original extraction pipeline).

|          | IN                  |    | C100   |    | C10    |    | DTD    |    | FMNIST |    | MNIST  |    |
|----------|---------------------|----|--------|----|--------|----|--------|----|--------|----|--------|----|
| model    | exc.                | r  | exc.   | r  | exc.   | r  | exc.   | r  | exc.   | r  | exc.   | r  |
| ViT-T    | +0.007 <sup>†</sup> | 1  | -0.008 | 15 | -0.051 | 20 | -0.024 | 20 | -0.012 | 11 | -0.041 | 19 |
| ViT-S    | -0.005 <sup>†</sup> | 18 | -0.005 | 15 | -0.090 | 20 | -0.035 | 20 | -0.060 | 20 | -0.051 | 20 |
| ViT-B    | -0.006 <sup>†</sup> | 20 | +0.001 | 12 | -0.083 | 20 | -0.042 | 20 | -0.069 | 20 | -0.041 | 20 |
| ViT-L    | -0.014 <sup>†</sup> | 20 | -0.029 | 20 | -0.096 | 20 | -0.054 | 20 | -0.079 | 20 | -0.060 | 20 |
| DINO-B   | -0.025              | 20 | -0.041 | 20 | -0.094 | 20 | -0.041 | 20 | -0.035 | 18 | -0.058 | 20 |
| DINOv2-S | -0.022              | 20 | -0.047 | 20 | -0.095 | 20 | -0.032 | 20 | -0.061 | 20 | -0.029 | 17 |
| DINOv2-B | -0.026              | 20 | -0.057 | 20 | -0.134 | 20 | -0.057 | 20 | -0.089 | 20 | -0.036 | 18 |
| DINOv2-L | -0.023              | 20 | -0.067 | 20 | -0.154 | 20 | -0.065 | 20 | -0.068 | 20 | -0.032 | 19 |
| DINOv2-G | -0.020              | 20 | -0.073 | 20 | -0.155 | 20 | -0.061 | 20 | -0.075 | 20 | -0.037 | 19 |
| CLIP-B   | -0.006              | 19 | -0.025 | 20 | -0.071 | 20 | -0.035 | 20 | -0.042 | 19 | -0.028 | 16 |
| CLIP-L   | -0.011              | 20 | -0.026 | 20 | -0.080 | 20 | -0.039 | 20 | -0.058 | 20 | -0.023 | 15 |
| SigLIP-B | -0.009              | 20 | -0.029 | 20 | -0.088 | 20 | -0.039 | 20 | -0.068 | 20 | -0.067 | 20 |

Table 31: The full vision p99.9 census with 20 spectrum-null replicates: excess and percentile rank r (number of the 20 null replicates above the real value; 20 = below every replicate). Agreement with the 5-replicate census on the 68 directly comparable cells: sign 64/68, significance at  $|z| \geq 2$  54/68; 56/72 cells lie below every replicate. <sup>†</sup>ImageNet rows for the supervised ViTs use the precomputed centroid store (precomputed centroid store, whose supervised-ViT centroids differ slightly from the census cache) and are not directly comparable to Table 6. The distributional statistic is stricter on supervised ViTs and stronger on DINO/DINOv2 than the supremum (Table 29).

| model    | Gauss. | IN<br>Haar | PC-perm. | Gauss. | C100<br>Haar | PC-perm. | Gauss. | C10<br>Haar | PC-perm. |
|----------|--------|------------|----------|--------|--------------|----------|--------|-------------|----------|
| ViT-T    | -0.005 | -0.014     | -0.038   | -0.044 | -0.031       | -0.055   | -0.057 | -0.027      | -0.050   |
| ViT-S    | -0.023 | -0.025     | -0.052   | -0.034 | -0.025       | -0.049   | -0.098 | -0.063      | -0.090   |
| ViT-B    | -0.027 | -0.025     | -0.054   | -0.018 | -0.011       | -0.039   | -0.090 | -0.054      | -0.078   |
| ViT-L    | -0.028 | -0.029     | -0.040   | -0.047 | -0.034       | -0.051   | -0.104 | -0.066      | -0.096   |
| DINO-B   | -0.005 | -0.002     | -0.008   | -0.068 | -0.056       | -0.068   | -0.100 | -0.070      | -0.097   |
| DINOv2-S | +0.010 | +0.015     | +0.013   | -0.050 | -0.029       | -0.051   | -0.093 | -0.083      | -0.098   |
| DINOv2-B | +0.004 | +0.015     | +0.006   | -0.042 | -0.007       | -0.041   | -0.132 | -0.120      | -0.124   |
| DINOv2-L | -0.013 | +0.000     | -0.011   | -0.053 | -0.008       | -0.044   | -0.153 | -0.136      | -0.122   |
| DINOv2-G | -0.000 | +0.016     | +0.001   | -0.080 | -0.032       | -0.074   | -0.155 | -0.138      | -0.110   |
| CLIP-B   | -0.028 | -0.028     | -0.048   | -0.046 | -0.036       | -0.053   | -0.079 | -0.045      | -0.075   |
| CLIP-L   | -0.016 | -0.017     | -0.029   | -0.042 | -0.031       | -0.052   | -0.087 | -0.049      | -0.082   |
| SigLIP-B | -0.019 | -0.020     | -0.044   | -0.042 | -0.032       | -0.046   | -0.096 | -0.056      | -0.081   |

Table 32: Excess under three constructions of the spectrum-matched null (10 replicates each): Gaussian coefficients (the paper’s), Haar-rotated coefficients (exact sample spectrum) and PC-permutation (exact per-component marginals). The Gaussian and PC-permutation constructions agree; the exact-sample-spectrum construction yields smaller excesses for low-effective-rank clouds on the small- $C$  sets (most visibly DINOv2 on CIFAR-100), so the size of those excesses is partly a property of the null hypothesis, while signs and the ImageNet readings are stable.

#### A.14 ORC EDGE TYPES

| model    | dataset | ORC within | ORC across | %neg within | %neg across |
|----------|---------|------------|------------|-------------|-------------|
| ViT-T    | IN      | 0.35       | 0.27       | 5.2         | 9.4         |
| ViT-T    | C100    | 0.51       | 0.42       | 1.4         | 2.7         |
| ViT-S    | IN      | 0.37       | 0.28       | 3.4         | 8.3         |
| ViT-S    | C100    | 0.61       | 0.48       | 0.0         | 3.0         |
| ViT-B    | IN      | 0.40       | 0.30       | 2.1         | 5.9         |
| ViT-B    | C100    | 0.61       | 0.48       | 0.0         | 1.7         |
| ViT-L    | IN      | 0.40       | 0.31       | 2.1         | 6.8         |
| ViT-L    | C100    | 0.61       | 0.47       | 0.6         | 2.1         |
| DINO-B   | IN      | 0.40       | 0.33       | 2.8         | 4.1         |
| DINO-B   | C100    | 0.60       | 0.45       | 0.0         | 1.4         |
| DINOv2-S | IN      | 0.37       | 0.33       | 1.8         | 1.2         |
| DINOv2-S | C100    | 0.60       | 0.44       | 0.0         | 0.8         |
| DINOv2-B | IN      | 0.58       | 0.64       | 0.3         | 0.1         |
| DINOv2-B | C100    | 0.66       | 0.54       | 0.0         | 0.1         |
| DINOv2-L | IN      | 0.69       | 0.72       | 0.0         | 0.0         |
| DINOv2-L | C100    | 0.70       | 0.64       | 0.0         | 0.1         |
| DINOv2-G | IN      | 0.70       | 0.74       | 0.0         | 0.0         |
| DINOv2-G | C100    | 0.77       | 0.70       | 0.0         | 0.0         |
| CLIP-B   | IN      | 0.36       | 0.29       | 4.2         | 7.9         |
| CLIP-B   | C100    | 0.60       | 0.46       | 0.0         | 3.1         |
| CLIP-L   | IN      | 0.35       | 0.27       | 4.0         | 8.4         |
| CLIP-L   | C100    | 0.57       | 0.43       | 0.0         | 2.2         |
| SigLIP-B | IN      | 0.34       | 0.26       | 5.0         | 7.5         |
| SigLIP-B | C100    | 0.59       | 0.43       | 0.0         | 2.3         |

Table 33: Edge-type ORC on the centroid kNN graph (30 WordNet superclasses): negative curvature concentrates on between-cluster bridges.

## A.15 PROMPT ROBUSTNESS

| model    | photo | name  | image | closeup | this is | wild  | def.  | gloss only | concept | discuss. |
|----------|-------|-------|-------|---------|---------|-------|-------|------------|---------|----------|
| BGE-base | 0.124 | 0.118 | 0.125 | 0.126   | 0.125   | 0.115 | 0.119 | 0.115      | 0.119   | 0.121    |
| E5-base  | 0.114 | 0.119 | 0.117 | 0.120   | 0.115   | 0.116 | 0.108 | 0.115      | 0.120   | 0.116    |
| GPT-2    | 0.131 | 0.145 | 0.110 | 0.086   | 0.080   | 0.089 | 0.105 | 0.124      | 0.112   | 0.105    |
| GPT-2-M  | 0.091 | 0.122 | 0.110 | 0.100   | 0.134   | 0.104 | 0.081 | 0.107      | 0.112   | 0.098    |

Table 34:  $\hat{\delta}$  across prompt templates (ImageNet class names; “gloss only” contains no class name).

| template   | GPT-2 S | GPT-2 M | GPT-2 L | GPT-2 XL |
|------------|---------|---------|---------|----------|
| photo      | −0.016* | −0.019* | −0.043* | −0.050*  |
| name_only  | −0.006  | −0.007  | −0.048* | −0.051*  |
| image      | −0.018* | −0.028* | −0.050* | −0.049*  |
| closeup    | +0.007  | −0.018* | −0.044* | −0.049*  |
| this_is    | −0.032* | −0.022* | −0.042* | −0.046*  |
| wild       | −0.023* | −0.067* | −0.014  | −0.010*  |
| def_pair   | −0.002  | −0.019* | −0.036* | −0.035*  |
| gloss_only | −0.020* | −0.030* | −0.033* | −0.036*  |
| concept    | −0.002  | −0.003  | −0.043* | −0.041*  |
| discussion | −0.010  | −0.013* | −0.041* | −0.040*  |

Table 35: Template-conditioned excess over the spectrum-matched null for the four GPT-2 sizes (10 templates; \*:  $|z| \geq 2$  against combined null and estimator noise). The scale trend of §4 is assessed on every template, not only on the paper’s baseline. Padding-free extraction (one prompt at a time, the canonical protocol of Table 12). Mean excess over templates: −0.012 (S), −0.023 (M), −0.040 (L), −0.041 (XL); cells at  $|z| \geq 2$ : 5, 8, 9 and 10 of 10. Only 3 null replicates per cell here, so the  $z$  are coarser than the census’s 20-replicate values: on the baseline template the census puts S and M within noise ( $z = -1.6/-1.2$ , excess −0.015/−0.010), so their starred cells are borderline, whereas the L/XL cells ( $z$  from −4 to −9) are not in doubt. Template sensitivity of raw  $\hat{\delta}$ : range up to 0.065 for causal LMs and  $\leq 0.012$  for embedders (prompt table in Appendix 34).

## A.16 SAMPLE-LEVEL FEATURES

| model    | CIFAR-100 (10/class) |         |      | DTD (22/class)     |         |      |
|----------|----------------------|---------|------|--------------------|---------|------|
|          | raw $\hat{\delta}$   | excess  | $z$  | raw $\hat{\delta}$ | excess  | $z$  |
| ViT-T    | 0.130                | −0.0151 | −1.7 | 0.124              | −0.0127 | −2.1 |
| ViT-S    | 0.121                | −0.0069 | −0.8 | 0.103              | −0.0165 | −2.4 |
| ViT-B    | 0.121                | −0.0075 | −0.9 | 0.092              | −0.0190 | −3.0 |
| ViT-L    | 0.100                | −0.0148 | −2.9 | 0.092              | −0.0108 | −1.2 |
| DINO-B   | 0.104                | −0.0106 | −1.7 | 0.110              | −0.0069 | −1.5 |
| DINOv2-S | 0.100                | −0.0091 | −1.3 | 0.145              | +0.0126 | +1.7 |
| DINOv2-B | 0.089                | −0.0063 | −1.1 | 0.139              | +0.0233 | +2.1 |
| DINOv2-L | 0.087                | −0.0068 | −1.5 | 0.131              | +0.0201 | +1.5 |
| DINOv2-G | 0.102                | +0.0048 | +0.4 | 0.136              | +0.0232 | +2.3 |
| CLIP-B   | 0.117                | −0.0176 | −2.3 | 0.125              | −0.0085 | −1.2 |
| CLIP-L   | 0.132                | +0.0022 | +0.2 | 0.120              | −0.0108 | −1.3 |
| SigLIP-B | 0.113                | −0.0194 | −2.3 | 0.114              | −0.0152 | −1.7 |

Table 36: The instrument transplanted to sample-level features, the object of prior latent-hyperbolicity readings ( $\approx 1000$  stratified training images per cell, spectrum-matched nulls). Raw sample-level  $\hat{\delta}$  sits in the same low band those works report, but the excess is mostly within null noise (6/24 cells at  $|z| \geq 2$ ; DINOv2 on DTD is sign-positive): raw sample-level readings are confounded exactly as raw centroid readings are, and the genuine tree excess of the census is a statement about *inter-class* geometry.

## A.17 TEXT-MODALITY TRANSFER

| model    | $\hat{\delta}$ | excess | trip. (cos) | NC H-R (pp) | FS H-R (pp) | CI95       |
|----------|----------------|--------|-------------|-------------|-------------|------------|
| BGE-base | 0.122          | -0.028 | 0.89        | -0.59       | +0.05       | $\pm 0.05$ |
| E5-base  | 0.127          | -0.024 | 0.89        | -0.43       | +0.04       | $\pm 0.03$ |
| GTE-base | 0.132          | -0.020 | 0.89        | -0.82       | +0.08       | $\pm 0.06$ |

Table 37: DBpedia Classes (219 leaf classes, 3 levels): genuine tree excess and strong sibling alignment, yet  $\hat{\delta}$  above the low- $\delta$  band correctly predicts marginal gains.

## A.18 HIERARCAPS MULTI-LEVEL HIERARCHY

| model    | $\rho(\text{level, radius})$ | % monotone | trip. L1 | trip. L2 | $\hat{\delta}$ leaves | excess |
|----------|------------------------------|------------|----------|----------|-----------------------|--------|
| BGE-base | $+0.80 \pm 0.31$             | 49.6       | 0.88     | 0.97     | 0.130                 | +0.003 |
| E5-base  | $+0.87 \pm 0.19$             | 53.9       | 0.90     | 0.97     | 0.139                 | +0.010 |
| GTE-base | $+0.47 \pm 0.55$             | 24.0       | 0.90     | 0.97     | 0.135                 | +0.006 |
| CLIP-B   | $+0.64 \pm 0.43$             | 27.2       | 0.86     | 0.95     | 0.123                 | -0.011 |
| CLIP-L   | $+0.51 \pm 0.49$             | 18.7       | 0.86     | 0.96     | 0.117                 | -0.010 |
| SigLIP-B | $+0.32 \pm 0.55$             | 9.8        | 0.83     | 0.91     | 0.138                 | +0.014 |

Table 38: HierarCaps test set (1000 four-level chains): radial ordering under the paper’s projection (chance for strict monotonicity =  $1/24 \approx 4.2\%$ ; shuffle controls  $\approx 0$ ), sibling triplet agreement (cosine; chance 0.5), and leaf  $\hat{\delta}$  with spectrum-null excess.

A.19 PARALLELOGRAM DEFECT  $\xi$ : A NORM-STRUCTURE DESCRIPTOR

For a triangle  $(a, b, c)$  with  $m$  the approximate midpoint of  $(b, c)$  (the minimizer of  $D(\cdot, b) + D(\cdot, c) + |D(\cdot, b) - D(\cdot, c)|$ ), the parallelogram defect is  $\xi = (D(a, m)^2 + \frac{1}{4}D(b, c)^2 - \frac{1}{2}D(a, b)^2 - \frac{1}{2}D(a, c)^2) / (2D(a, m) \text{diam})$ , averaged over  $2 \times 10^4$  sampled triangles: 0 in Euclidean space, positive on spheres, negative in hyperbolic space and on trees.

| space / model                   | $\xi$ (mean) | frac. negative | null $\xi$         | excess ( $\pm$ CI95) | $z$ |
|---------------------------------|--------------|----------------|--------------------|----------------------|-----|
| iid Gaussian ( $d=768$ )        | +0.092       | 0%             | —                  | —                    | —   |
| $S^d$ (geodesic)                | +0.113       | 0%             | —                  | —                    | —   |
| $S^d$ (chord)                   | +0.115       | 0%             | —                  | —                    | —   |
| $\mathbb{H}^2$ region ( $R=6$ ) | -0.194       | 96%            | —                  | —                    | —   |
| balanced tree (depth 9)         | -0.239       | 96%            | —                  | —                    | —   |
| ViT-T                           | +0.027       | 25%            | $+0.027 \pm 0.002$ | $-0.001 \pm 0.003$   | -0  |
| ViT-S                           | +0.027       | 23%            | $+0.042 \pm 0.002$ | $-0.015 \pm 0.005$   | -7  |
| ViT-B                           | +0.033       | 16%            | $+0.047 \pm 0.002$ | $-0.014 \pm 0.004$   | -6  |
| ViT-L                           | +0.028       | 21%            | $+0.046 \pm 0.002$ | $-0.018 \pm 0.005$   | -7  |
| DINO-B                          | -0.008       | 55%            | $+0.052 \pm 0.003$ | $-0.059 \pm 0.005$   | -21 |
| DINOv2-S                        | -0.020       | 66%            | $+0.061 \pm 0.004$ | $-0.081 \pm 0.008$   | -20 |
| DINOv2-B                        | -0.049       | 83%            | $+0.075 \pm 0.003$ | $-0.123 \pm 0.005$   | -45 |
| DINOv2-L                        | -0.068       | 90%            | $+0.080 \pm 0.002$ | $-0.149 \pm 0.004$   | -66 |
| DINOv2-G                        | -0.070       | 89%            | $+0.082 \pm 0.002$ | $-0.152 \pm 0.004$   | -69 |
| CLIP-B                          | +0.019       | 32%            | $+0.024 \pm 0.002$ | $-0.005 \pm 0.003$   | -3  |
| CLIP-L                          | +0.028       | 28%            | $+0.032 \pm 0.002$ | $-0.004 \pm 0.004$   | -2  |
| SigLIP-B                        | +0.036       | 20%            | $+0.031 \pm 0.002$ | $+0.005 \pm 0.004$   | +2  |

Table 39: Parallelogram curvature-sign estimator  $\xi$  on reference geometries (top) and ImageNet centroids (bottom), with the spectrum-matched null, excess and  $z$ -score for each model (20 null replicates; noise combines null and estimator s.d.): matched nulls have positive  $\xi$ , so raw readings understate curvature.

| model    | $\xi_{\text{geo}}$ | frac. neg. | excess  | $z$   |
|----------|--------------------|------------|---------|-------|
| ViT-T    | +0.031             | 0.24       | −0.0048 | −2.7  |
| ViT-S    | +0.065             | 0.08       | +0.0092 | +6.2  |
| ViT-B    | +0.067             | 0.07       | +0.0144 | +10.5 |
| ViT-L    | +0.074             | 0.04       | +0.0113 | +7.9  |
| DINO-B   | +0.085             | 0.05       | +0.0014 | +1.0  |
| DINOv2-S | +0.100             | 0.02       | +0.0033 | +2.3  |
| DINOv2-B | +0.108             | 0.01       | +0.0046 | +4.9  |
| DINOv2-L | +0.108             | 0.01       | +0.0040 | +3.9  |
| DINOv2-G | +0.094             | 0.01       | −0.0114 | −12.5 |
| CLIP-B   | +0.025             | 0.28       | −0.0129 | −7.1  |
| CLIP-L   | +0.033             | 0.25       | −0.0120 | −4.8  |
| SigLIP-B | +0.051             | 0.14       | +0.0025 | +1.2  |

Table 40:  $\xi$  on the angular geometry: centroids L2-normalized, distances the spherical geodesic, spectrum null built on the normalized cloud and re-normalized (20 replicates; ImageNet centroid store). Probes whether the deep negative  $\xi$ -excess of DINOv2 on ImageNet (Table 39) is a norm-structure effect. Outcome: it is for S/B/L, whose angular  $\xi$ -excess turns sphere-leaning positive, while DINOv2-G and CLIP-B/L remain negative on the sphere: the Euclidean  $\xi$  signal of DINOv2 is largely carried by norm structure.

| reference (no curvature unless stated)     | $\xi$   | frac. negative |
|--------------------------------------------|---------|----------------|
| iid Gaussian                               | +0.0919 | 0.00           |
| Gaussian, lognormal radii ( $\sigma=0.3$ ) | −0.0621 | 0.99           |
| Gaussian, lognormal radii ( $\sigma=0.6$ ) | −0.0607 | 1.00           |
| Gaussian, lognormal radii ( $\sigma=1.0$ ) | −0.0496 | 0.99           |
| core + shell mixture                       | −0.3217 | 0.79           |
| uniform sphere                             | +0.1154 | 0.00           |
| sphere with radial jitter                  | −0.0571 | 1.00           |

Table 41:  $\xi$  on references with heterogeneous norms but no curvature ( $n=1000$ ,  $d=768$ , 5 seeds). Norm dispersion alone drives  $\xi$  negative (lognormal radii, radial jitter on a sphere, core+shell): the estimator reads norm structure, not curvature, and the paper reports it only as such.

## B ILLUSTRATIVE VISUALIZATION

Figure 9 is a low-dimensional illustration of the hub-and-spoke organization; every quantitative comparison in the paper is made in the full feature space (§4).

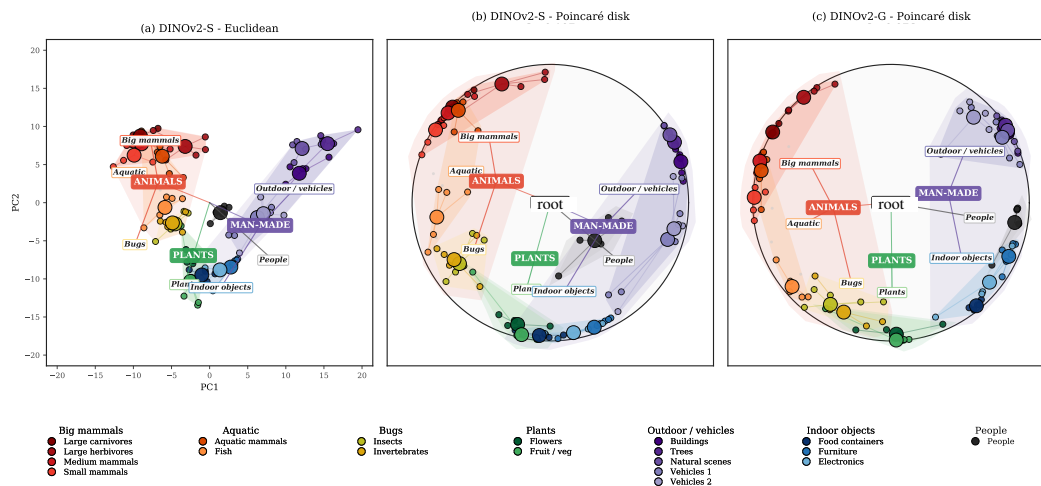


Figure 9: Illustrative 2-D projection of the 100 CIFAR-100 class centroids (per-panel PCA fit on the 20 super-class hubs; panels preserve 22–30% of hub variance and 6–10% of spoke length, so visual compactness is not comparable across panels and quantitative comparisons are made in full dimension). (a) DINOv2-S, Euclidean; (b) DINOv2-S after the projection of §3; (c) DINOv2-G.